EIDO Incident Transfer Acceptance Test Specification

Transfer of an Active Incident Between Two Public Safety Answering Points Using the NENA Emergency Incident Data Object

Version 0.1, Draft for External Review

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Document Control

Version	Date	Change	Author
0.1 (draft)	9 November 2026	First version for external review. Part A baseline aligned to NENA-STA-024.1.1-2025 Push transport; Part B with eleven test cases in three suites, each written as a full test description with numbered steps and verdict conditions; three fixtures; static checks executed on the author's computer on 8 November 2026 (Appendix A). Operational cases not executed.	Jungeun Kim

How to Read This Document

This document can be read without the NENA standards or the JSON files at hand. The scenario is described in plain words first; the standards are cited so that a reader can check each statement, not because they are needed to follow the argument.

The scenario in one paragraph. A 9-1-1 center (PSAP A) has taken a call about a traffic accident and has opened an incident. The scene turns out to be in the area served by a neighboring center (PSAP B). PSAP A asks PSAP B to take over the incident by sending it a structured data message, the EIDO, which carries the incident type, status, location, the agencies involved, and the call details. PSAP B must check that the message is complete and valid, answer PSAP A with a response code, create its own incident record from the message, and show it to a dispatcher. This document says what PSAP B should check at each of those steps, and how to decide whether each check passed.

Reading order. A reader who wants the argument reads Sections 1, 2, and 5 (about six pages). A reader who will execute or review the tests reads Part A (Section 3) for the exchange being tested, then Part B (Section 4): 4.3 is a one-page index of the eleven cases and 4.4 gives each case in full, with numbered steps and verdict conditions. The appendices hold the raw material: the check reports, the fixtures, the element record, and the script.

Abbreviation	Meaning
PSAP	Public Safety Answering Point: a 9-1-1 call center
NG9-1-1	Next Generation 9-1-1: the IP-based 9-1-1 architecture defined by NENA
NENA	National Emergency Number Association, the body that publishes the 9-1-1 standards cited here
EIDO	Emergency Incident Data Object: the JSON message that carries an incident's current state (NENA-STA-021)
STA-021, STA-024	NENA-STA-021.1a-2022 (EIDO content) and NENA-STA-024.1.1-2025 (EIDO conveyance, that is, how the EIDO is sent)
FE	Functional element: a software component of the NG9-1-1 architecture with a defined interface
Logger FE	The functional element that stores log events from other elements
CAD	Computer-aided dispatch: the system a dispatcher uses to manage incidents and units
IUT	Implementation under test: the system whose behavior the test judges (PSAP B's system

	in most cases)
IXIT	Implementation extra information for testing: configuration facts the tester needs from the IUT's operator (for example, its incident type mapping and its service area)
JSON, JSON Schema, OpenAPI	The text format of the EIDO; the language in which its structure is defined; the file format (openapi.yaml) in which NENA publishes that definition
Registry	A NENA-maintained list of allowed coded values (for example, the list of incident types)
PIDF-LO	The XML format for a location carried inside an EIDO
RFC 3339	The Internet standard for writing dates and times (for example, 2021-04-30T14:45:00.000-04:00)
TLS, PCA	Transport Layer Security; PSAP Credentialing Agency, the certificate authority for NG9-1-1 systems
MiB	Mebibyte, 1,048,576 bytes
ICE	NENA Industry Collaboration Event: a vendor interoperability test event
CACN	Texas A&M Cybersecurity and Communications Network (operator of an NG9-1-1 testing program)
IPAWS	Integrated Public Alert and Warning System (FEMA)

Notation	Meaning
TD-01, TD-02R, TD-02A	The three test fixtures (prepared EIDO files). TD-01 is NENA's published sample; TD-02R is the request as sent; TD-02A is the state after acceptance. Section 3.5.
TC-01 to TC-11	The eleven test cases. Section 4.3 (index) and 4.4 (full descriptions).
Suite 1, 2, 3	Groups of cases: schema conformance; document rules; operational acceptance. Section 4.1.
Level 1, Level 2	Static checks on the file alone; checks of the receiving system's behavior. Section 4.1.
R1 to R12	Rule checks: constraints stated in the standards' text that the schema does not enforce. Appendix D.1.
M1 to M10	Mutations: deliberately broken copies of a fixture used to show which defects each check reports. Appendix D.1.
S1 to S7	Steps of the transfer sequence. Section 3.2.
t_send, t_recv, t_ack, t_process, t_display; d_ack, d_use	Time points and intervals. Section 4.2.
PASS, FAIL, NA, BLOCKED, OBSERVE	Verdicts. Section 4.1.
Standard / Proposed	Whether an expected result follows from a normative statement (with clause) or is the author's proposed criterion.
$id, $ref	The identifier of an object inside an EIDO, and a pointer to such an identifier.
requestId, responseCode	The fields of the request and response messages of STA-024 section 3.8.

1. Purpose and Scope

This document specifies how a receiving Public Safety Answering Point (PSAP) can verify, against written acceptance criteria, that an active incident transferred to it from another PSAP has arrived complete, current, and usable. The exchange format is the NENA Emergency Incident Data Object (EIDO), a JSON document defined in NENA-STA-021 [1]. The transport is the Push transport of NENA-STA-024.1.1-2025 [3]. The document has two parts. Part A fixes the exchange boundary, the transfer sequence, the data elements under test, the error and response conditions, and the test fixtures. Part B defines the test cases, their verdict rules, the measurements, and the environments in which each case can be executed.

What this document is not. It is not a certification procedure and does not produce a certificate or a declaration of conformity. It is not a proposal to change the EIDO standard. It does not evaluate any product. It makes no claim about any real agency; all fixtures are synthetic.

Scope of this version. One scenario is covered: PSAP A, which received the call and owns the incident, requests that PSAP B take over the incident, and PSAP B accepts. Eleven test cases are defined in three suites: schema conformance, document rules, and operational acceptance. Resource requests, situation reporting to an Emergency Operations Center, split and merge of incidents, and callbacks are outside this version.

Status. The static checks that support Suite 1 and Suite 2 were executed by the author on three fixtures on 8 November 2026 (Section 5, Appendix A). No operational test case (Suite 3) has been executed. Where a threshold is needed, it is marked "proposed threshold, subject to validation".

1.3 What this document contributes, and for whom

Three things in this document did not exist in public form before it, as far as the author has been able to establish (Section 2.2).

- A documented and reproducible finding about the published EIDO schema: of ten deliberately introduced defects, the schema reports four and misses six (values outside three registries, a malformed timestamp, both location forms present, a dangling reference, two owning agencies). The checks that catch the six are written down and the script is included (Appendix D). This is small, concrete, and verifiable by anyone with the three public files.
- A set of receiving-side acceptance tests for an incident transfer, each with numbered steps and verdict conditions, and each expected result tagged as following from a standard or proposed by the author. The existing programs test functional elements or vendor interoperability; none publishes what a receiving agency should check.
- Two constructed fixtures (TD-02R, TD-02A) that represent the two states of a transfer, because the one published sample represents a call transfer without an incident component and cannot be used for this purpose.

Who might use it. The groups that run NG9-1-1 test programs (the Texas A&M program, the organizers of NENA ICE events), the NENA working group that maintains the EIDO schema (for the first item), and researchers in public safety data exchange. Agencies will use a document of this kind only through such programs or through their vendors, and only once EIDO transfers between agencies are in production, which in most of the country they are not. The author does not claim that the document will be widely adopted; the claim is that it fills a specific gap, that its results can be reproduced, and that the next step is review by the groups named above.

2. Background

2.1 NG9-1-1 and the EIDO

In Next Generation 9-1-1 (NG9-1-1), incident information moves between functional elements and between agencies as an EIDO. NENA-STA-021 defines the content of the EIDO: a set of data components (agency, agent, call, incident, location, additional data, and others), the elements of each component, and the registries from which coded values are drawn. The standard states that an EIDO conveys the current state of an incident, not its history, and that only one agency component in an EIDO may be marked as the owning agency. NENA-STA-024 defines how an EIDO is moved: by subscription and notification, by direct request to a dereference service, inside SIP call signaling, or by an unsolicited Push to an agency that has no subscription. The Push transport is the one that matches a transfer of an active incident from one PSAP to another PSAP; its RequestForService command lists "Primary PSAP to an out of area primary PSAP" among its use cases.

2.2 Existing testing programs and the gap addressed here

Three publicly documented programs test NG9-1-1 implementations. None of them publishes test descriptions for the acceptance of a transferred EIDO at the receiving agency.

Program	What it tests	Public material	Relation to this document
NENA Industry Collaboration Events (ICE 12, October 2024; ICE 13, October 2025)	Interoperability of participating vendors' implementations, including EIDO exchange and PSAP to PSAP transfer	Summary reports; detailed test cases and results are for participants	Tests that vendors can exchange EIDOs. Does not give a receiving agency a written acceptance procedure.
Texas A&M CACN NG9-1-1 Interoperability Testing Program (repository tamu-edu/ng911-dev, v5.6 release, 12 October 2026, MIT License) [6]	Conformance of i3 functional elements (BCF, ESRP, ECRF, LVF, LIS, Bridge, Logging, Policy Store, CHFE) by stimulus and response	Test description template, 104 current test descriptions across nine folders, automated test suite	Defines the test description format reused in Part B (interfaces marked active or passive; preamble, body, postamble; verdict rules). In the public repository the EIDO appears once, as a logged event type; there is no EIDO conveyance test description.
FEMA IPAWS Guide for Independent Testing of EAS Equipment (June 2012) [7]	CAP v1.2, IPAWS Profile, and ECIG conformance of alerting equipment through accredited independent test authorities	Public guide with test suites and case identifiers	Alerting, not 9-1-1. Reused for its structure: numbered suites, pass or fail versus observe cases, a test report separate from any declaration, and an explicit statement that testing is not certification.

A 2017 study by the Institute for Defense Analyses (IDA Document D-8728) [8] compared the NIEM 4.0 emergency management domain with the EIDO's predecessor, the Emergency Incident Data Document, and reported the elements that were missing or only partially covered. That study is a data model gap analysis. The element table in Section 3.3 of this document is work of the same kind, limited to the elements that a transfer exercises; the test cases in Part B are the part that the earlier study did not attempt.

The gap this document addresses is therefore narrow: a receiving PSAP has a standard that tells it what a valid EIDO is and a standard that tells it how the EIDO arrives, but no published set of acceptance tests that tells it what to check, in what order, with what verdict, when an incident it did not create arrives from another jurisdiction.

2.3 How this document was built

The element names, cardinalities, registries, and normative statements were taken from the standards' texts. The schema behavior was established by validating fixtures against the published openapi.yaml with a JSON Schema validator and by deliberately breaking the fixtures to see which defects the schema reports and which it does not. The rules that the schema does not enforce were written as explicit checks. The result is that every expected result in Part B is tagged either "Standard" (follows from a normative statement or from the schema) or "Proposed" (an acceptance criterion the author proposes, not stated in any standard).

3. Part A. Data Exchange Baseline

3.1 Exchange boundary

Interface or party	Role in this version	Mode in test	Notes
PSAP A (sending agency)	Received the call; owns the incident at the start; issues the EIDO and sends a	Active (stimulus source)	Fixture issuer: issuingElementIdentification

	RequestForService		idx.state.pa.us (synthetic)
PSAP B (receiving agency)	Implementation under test. Receives the Push, validates, responds with ResponseToRequest, creates its incident record on 200	Active (response) and Passive (record, display observed)	Synthetic agency $id psap-b.example.org
Push transport (NENA-STA-024 section 3.8)	WebSocket text frames, UTF-8, TLS. Commands requestForService and responseToRequest	Passive (monitored); Active only in the emulated testbed	Credentials traceable to the PSAP Credentialing Agency are assumed, not tested
Logger FE (NENA-STA-024 section 3.14)	Receives an EidoLogEvent from every FE that sends or receives an EIDO	Passive (observation point)	direction incoming or outgoing; requestId carried
Static check environment	A single computer with the published schema, the fixtures, and the check script	Active, no network	The only environment in which this version was executed
Out of scope	ESInet routing, call audio, subscription and notification transport, dereference service, EOC display, resource messaging, split and merge, callbacks		

Mode follows the convention of the Texas A&M test descriptions: an interface is Active when the test system sends to it or receives from it as part of the stimulus or response, and Passive when it is only monitored.

3.2 Transfer sequence and observation points

The sequence follows the RequestForService command of NENA-STA-024 section 3.8.1. Observation points are the places where Part B reads a value or a time.

Step	Action	Actor	Observation point	Symbol	Basis
S1	PSAP A sets the incident state for transfer: PSAP B added to agencyComponent with role TransferredTo; PSAP A remains the owning agency; incidentComponent present	PSAP A	Content of the EIDO (fixture TD-02R)		STA-021 sections 2.6, 2.9; STA-024 3.8.1: the EIDO in a RequestForService MUST include an incident data component
S2	PSAP A sends requestForService {requestId, eido} over the Push transport and logs an EidoLogEvent (outgoing)	PSAP A	PSAP A send log; EidoLogEvent outgoing with requestId	t_send	STA-024 3.8.1; 3.14
S3	PSAP B receives the frame and logs an EidoLogEvent (incoming, same requestId)	PSAP B	Logger FE: EidoLogEvent incoming	t_recv	STA-024 3.14: any FE that receives an EIDO MUST log it
S4	PSAP B validates the EIDO: schema, registries, document rules	PSAP B	Validation outcome and, on failure, the element name	v_result	STA-021 content rules; STA-024 3.13 discrepancy categories NonConformant, BadDomain, EIDOMissingElement
S5	PSAP B returns responseToRequest {requestId, responseCode, responseText}	PSAP B	Response frame received by PSAP A; code value	t_ack, code	STA-024 3.8.1 ResponseToRequest; no response within the local timeout is logged with status 408
S6	On code 200, PSAP B creates an active incident under the same Incident Tracking Identifier	PSAP B	PSAP B incident record: $id, type, status, location, owning agency	t_process	STA-024 3.8.1: a recipient that responds 200 MUST create an active incident based on the EIDO
S7	The record is presented to the PSAP B operator; PSAP B may then publish its own EIDO marking itself as owner (state TD-02A) and either side may subscribe to the other's incident	PSAP B	Operator display; PSAP B outgoing EIDO	t_display	STA-024 3.8.1 notes (subscription after 200 is MAY); STA-021 2.6 (once true, only the agency that set it should clear it)

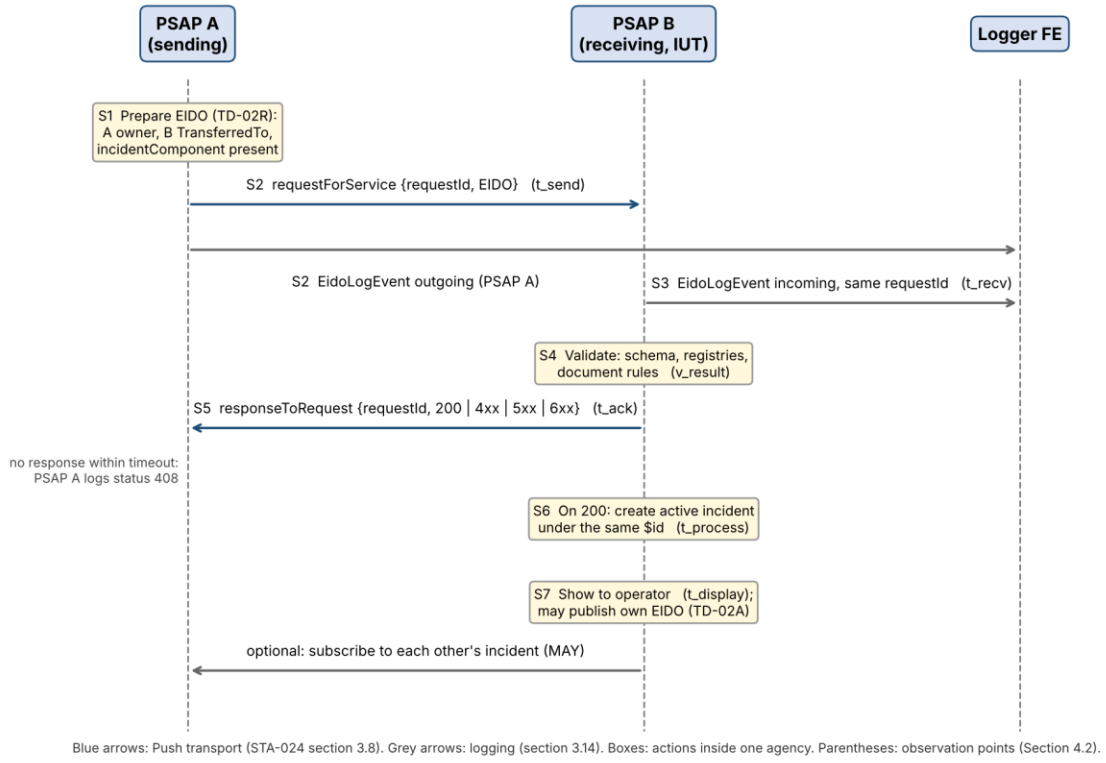


Figure 1. Transfer sequence S1 to S7 with the observation points of Section 4.2.

Two fixture states correspond to this sequence. TD-02R is the EIDO as sent at S2: PSAP A owner, PSAP B TransferredTo, every component last updated by PSAP A. TD-02A is the state after S7: PSAP B owner. The standard does not state who clears the sender's indicator or in what order; this version assumes that PSAP A clears its own indicator after receiving code 200, and records this as an assumption, not a requirement.

3.3 Data elements under test

Twenty-two elements are selected from the thirty-five element working record (Appendix C) because the transfer test cases exercise them. "Schema enforced" says what the published openapi.yaml checks: Value Y means an enumerated value is checked; Presence Y means the property is in a required list, which applies only when its parent component is present; N means the schema does not require or constrain the element, which does not by itself make an absence a nonconformance. A rule check in Appendix D covers the N rows that carry a document rule.

Element / path	STA-021 clause	OpenAPI type	Schema enforced	In TD-01	In TD-02R	Meaning in the transfer test	Cases
EIDO.$id	2.1, 2.4	EIDO + PrologueType	Presence Y	Y	Y	Global Incident Tracking Identifier; the same incident must be recognizable at PSAP B by this value	TC-01, 06, 07
EIDO.lastUpdateTimeStamp	2.1, 2.4	EIDO + PrologueType	Presence Y (format N)	Y	Y	EIDO-level update time; RFC 3339 form checked by rule R4	TC-01
EIDO.updatedByAgencyReference	2.1, 2.4	PrologueType > ReferenceType	Presence Y	Y	Y	Agency that produced the conveyed state; $ref resolution checked by rule R2	TC-01
EIDO.updatedByAgentReference	2.1	PrologueType > ReferenceType	N	Y	Y	Agent trace; conditional in the text	TC-01
EIDO.eidoVersion	2.4	EmergencyIncidentDataObjectType	Value Y (enum 1.0)	Y	Y	Version check before parsing	TC-01
EIDO.issuingElementIdentification	2.4	EmergencyIncidentDataObjectType	Presence Y	Y	Y	Issuing functional element; distinct from the receiving PSAP	TC-01
EIDO.incidentComponent	2.4, 2.9	IncidentInformationType	N	N	Y	Carries incident facts; required by STA-024 3.8.1 in a RequestForService	TC-01, 02, 03
incidentComponent.incidentTypeCommonRegistryText	2.9, 3.2	IncidentInformationType	Presence Y (value N)	N	Y (MVA)	Common incident type; registry value checked by rule R11	TC-02, 03
incidentComponent.incidentStatusCommonRegistryText	2.9, 3.3	IncidentInformationType	N	N	Y (Active)	Incident state; distinct from call state; registry checked by rule R7	TC-06, 08
incidentComponent.internalIncidentId	2.9	IncidentInformationType	N	N	Y	PSAP A local number kept for cross reference; not the tracking identifier	TC-08
incidentComponent.locationReference	2.9	IncidentInformationType > ReferenceType	N	N	Y	Links the incident to its scene location	TC-05, 08
incidentComponent.lastUpdateTimeStamp	2.1, 2.9	IncidentInformationType + PrologueType	Presence Y	N	Y	Freshness of the incident component; the comparison value in TC-06	TC-06
callComponent[].$id	2.1, 2.10	CallInformationType + PrologueType	Presence Y	Y	Y	Call Identifier; not the Incident Tracking Identifier	TC-01
callComponent[].direction	2.10	CallInformationType	Value Y	Y	Y	Call direction enum	TC-01
callComponent[].callStateRegistryText	2.10	CallInformationType	Presence Y	Y	Y	Call state; not incident status	TC-01
EIDO.locationComponent	2.4, 2.18	LocationInformationType	N	Y (2)	Y (3)	Location objects; by-value or by-reference, one and only one (rule R8)	TC-05
locationComponent[].locationTypeDescriptionRegistryText	2.18, 3.8	LocationInformationType	Presence Y (value N)	Y	Y	Distinguishes CurrentIncident from Caller and RoutingLocation; registry checked by rule R6	TC-05, 08
EIDO.agencyComponent	2.4, 2.6	AgencyType	Presence Y	Y (1)	Y (2)	Both agencies must appear in a transfer	TC-01, 04
agencyComponent[].$id	2.1, 2.6	AgencyType + PrologueType	Presence Y	Y	Y	Agency Identifier	TC-01
agencyComponent[].agencyRoleDescriptionRegistryText	2.6, 3.1	AgencyType	Presence Y (value N)	Y	Y	CallReceiving versus TransferredTo; registry checked by rule R5	TC-01, 08
agencyComponent[].agencyType	2.6	AgencyType	Presence Y	Y	Y	Agency classification (psap)	TC-01
agencyComponent[].incidentOwningAgencyIndicator	2.6	AgencyType	N	N	Y	At most one true (rule R10); the handover is the operational core of the transfer	TC-04, 08

Non-equivalences that the test cases rely on: EIDO.$id is not callComponent[].$id and is not an agency's internal incident number. Call state is not incident status. The EIDO-IncidentStatus-Common values are not a one-to-one mapping of ICS Form 210 resource status. Operational CAD field mappings are proposed until a receiving system and its data dictionary are specified.

3.4 Error and response conditions

The response codes come from the ResponseToRequest status table for RequestForService in NENA-STA-024 section 3.8.1. The sending side action is the one given in that table's notes; where the table gives none, the action is marked Proposed.

Code	Meaning (STA-024)	Sending side action	Receiving side expectation in this version	Basis
180 / 182	Offered to agents / Queued	Wait for a final code	Final code follows	Standard
200	OK, request accepted	Follow with a subscription to the recipient's incident (should); clear own owning indicator (assumption, Section 3.2)	MUST create an active incident based on the EIDO	Standard (creation); Proposed (indicator)
302	Moved temporarily	Resubmit to the location given		Standard
403	Forbidden, policy non-compliant	Do not retry unchanged; notify operator	Log; optional EIDOConveyance discrepancy report NotAccepted by sender	Standard (code); Proposed (operator notice)
404	Recipient does not provide for the request	Find an alternate agency		Standard
408	Request timeout	Log with status 408 when no response is received		Standard (3.14)
488	Not acceptable here: dispatch type or resources not provided	Read the reason; find an alternate agency		Standard
500 to 505	Server errors; 504 server time-out	Find an alternate agency; for 504, retry or alternate per policy		Standard
513	Message too large (recipient maximum MUST be at least 10 mebibytes)	Reduce the EIDO or find an alternate agency	Reject with 513 only above its maximum; never below 10 MiB	Standard
603	Decline, request cannot be fulfilled	Find an alternate agency; notify operator		Standard (code); Proposed (notice)
606	Not acceptable: agency does not serve the requested area	Notify operator; re-check location and jurisdiction	Return 606 when the incident location is outside its service area, or accept and flag for operator review (TC-05)	Standard (code); Proposed (behavior)

Content defects found at S4 map to the discrepancy categories of NENA-STA-024 section 3.13, which an entity MAY report: NonConformant (schema failure), BadDomain (value outside the registry or domain), EIDOMissingElement (an expected element absent), EIDOInconsistent. The rule identifiers of Appendix D are related to these categories in Section 4.3. Duplicate delivery and out-of-order updates are not addressed by either standard; the requestId returned in every response is the only standard means of relating a response to a request, and the handling in TC-06 and TC-07 is therefore Proposed.

3.5 Test fixtures and provenance

Three fixtures are used. All content is synthetic. No real 9-1-1 data, agency, delivery, acceptance, or ownership transition is represented.

Fixture	File	State represented	SHA-256
TD-01	Sample call transfer EIDO.json (NENA911/EIDO-JSON repository, unaltered)	Published call transfer sample. One agency, no incidentComponent, no owning indicator. Baseline only; does not satisfy the preconditions of an active incident transfer.	`4f98a553e6e095e25c96a377a068c18ea87fe1674c65199f9a2f7afe0d5bf57d`
TD-02R	TD-02R_request_payload.json (author constructed)	Request payload at S2: PSAP A owner (true), PSAP B TransferredTo (false), all components last updated by PSAP A; incidentComponent MVA, Active; CurrentIncident location added	`c04a620e092a66a59d481b9799d8b2358f10b02cc96a41c5809aa4b24a601e58`
TD-02A	TD-02_author_constructed_incident_transfer.json (author constructed)	State after acceptance (S7): PSAP B owner (true), PSAP A false, PSAP B component updated by PSAP B. Issuer identification not changed; see limitation in Section 7.	`c49537b16b1e650d14b3fde55c200b36bc56f22e9fcc01d114540b79a7898ca2`

Differences between the fixtures are listed in Appendix B. The CurrentIncident location in TD-02R and TD-02A reuses the PIDF-LO civic address of the sample's RoutingLocation; the coordinates are not independently verified and the address is in a different state from the synthetic agency identifiers, which is acceptable for a synthetic fixture and is noted so that TC-05 variants change the address rather than reuse it. The static check results for each fixture are in Section 5 and Appendix A.

4. Part B. Test Cases

4.1 Levels, suites, and verdicts

Level	Definition	Executed in this version
Level 1. Static conformance	The EIDO is valid against the published schema (1a) and against the document rules of the standard that the schema does not enforce (1b): registry values, RFC 3339 form, by-value versus by-reference exclusivity, single owning agency, $ref resolution, embedded XML well-formedness, size.	Yes, on three fixtures, by the author (Appendix A)
Level 2. Operational acceptance	The EIDO is accepted by the receiving agency's system and operator in the way the standards require or this document proposes: record creation on 200, no stale overwrite, no duplicate record, ownership conflict flagged, jurisdiction check, operator can act on the record, delivery and display times recorded.	No

Suite	Cases	Content
Suite 1. Schema conformance	TC-01, TC-02, TC-10	Required elements, enumerations, size limit
Suite 2. Document rules	TC-03, TC-04	Registry values and single ownership, which the schema does not enforce
Suite 3. Operational acceptance	TC-05, TC-06, TC-07, TC-08, TC-09, TC-11	Jurisdiction, freshness, duplicates, operator usability, timing, response handling

Verdicts. PASS and FAIL are assigned by the rule stated in each case. NA means the case does not apply to the fixture (for example, a registry rule on an EIDO without an incident component). BLOCKED means a precondition could not be met in the environment. OBSERVE is recorded for a case that is executed for information only in a given environment and does not count toward a conformance statement; the distinction between pass or fail cases and observe cases is taken from the FEMA IPAWS independent testing guide. Each case is structured as preamble (preconditions and fixture preparation), body (stimulus, expected response, verdict), and postamble (restore the receiving record to its pre-test state), following the Texas A&M test description template.

4.2 Measurements

Symbol	Definition	Where recorded	Standard object
t_send	Time PSAP A sends requestForService	PSAP A send log	EidoLogEvent, direction outgoing, requestId
t_recv	Time PSAP B logs the incoming EIDO	Logger FE	EidoLogEvent, direction incoming, same requestId
t_ack	Time PSAP A receives responseToRequest	PSAP A receive log	ResponseToRequest {requestId, responseCode, responseText}
t_process	Time PSAP B's incident record is created or updated	PSAP B record	
t_display	Time the record is presented to the PSAP B operator	PSAP B display log	
d_ack = t_ack - t_send	Delivery and acknowledgment time	Computed	
d_use = t_display - t_send	Time to operator availability	Computed	
v_result	Validation outcome; on failure the element name and the rule or	PSAP B validation log	Discrepancy category of STA-024 3.13 where reported

	schema path		
code	ResponseToRequest responseCode	PSAP A receive log	STA-024 3.8.1 status table

Version 0.1 records d_ack and d_use without pass or fail thresholds. Thresholds will be proposed after consultation with practitioners and marked "proposed threshold, subject to validation" until confirmed in execution. Clock synchronization between the two systems is a precondition for d_ack and d_use and is recorded with the result.

4.3 Test case index

One row per case. Section 4.4 gives each case in full with numbered steps and verdict conditions; this table is for orientation and for the results sheet.

Basis: "Standard" means the expected result follows from a normative statement of NENA-STA-021 or NENA-STA-024 or from the published schema (clause given); "Proposed" means the author proposes the criterion for operational acceptance. Env: S = static check on one computer (executed); E = emulated testbed with sender and receiver emulators (not yet built); L = live systems of two agencies (requires agreements). RQ refers to rows of the element table in Section 3.3 by element name.

ID	Suite / Lvl	Preamble (precondition, fixture)	Stimulus	Expected response	Verdict	Observation / measure	Basis (clause)	Env	Postamble
TC-01	1 / L1 and L2	Push session available (E, L) or static environment (S). Fixture TD-02R unaltered.	Send requestForService {requestId, TD-02R}. In S: run the static check on TD-02R.	L1: schema valid; rules R1 to R12 PASS. L2: responseToRequest code 200 with the same requestId; EidoLogEvent incoming logged; PSAP B record created under the same $id with type MVA, status Active, CurrentIncident location, and PSAP A shown as owner.	L1: PASS if zero schema errors and no rule FAIL; FAIL any other case. L2: PASS if code 200 and the record exists with the four fields equal to the sent values; FAIL any other case.	v_result; EidoLogEvent; record; d_ack, d_use	Standard. STA-021 2.4, 2.6, 2.9, 2.18; schema required lists; STA-024 3.8.1 (200 MUST create an active incident), 3.14 (MUST log)	S (L1 executed, PASS); E, L (L2)	Delete the PSAP B record created by the test
TC-02	1 / L1	As TC-01.	TD-02R with incidentComponent.incidentTypeCommonRegistryText removed.	Schema error naming incidentTypeCommonRegistryText under incidentComponent; no record created; in E or L a response code other than 200 and, if reported, discrepancy EIDOMissingElement or NonConformant.	PASS if the error names the element and no record is created; FAIL any other case.	v_result; record count	Standard. STA-021 2.9 (required when the component is present); openapi.yaml IncidentInformationType required	S (schema detects: executed as a mutation of the same kind, M1, M3, M9); E, L	None if no record was created
TC-03	2 / L1	As TC-01.	TD-02R with incidentTypeCommonRegistryText set to INVALID_CODE (mutation M10). Variant: agencyRoleDescriptionRegistryText set to Transferring (M4).	Schema reports no error; rule R11 (or R5) FAIL naming the value; no record created or record flagged for review; if reported, discrepancy BadDomain.	PASS if the rule names the element and the value and the record is not created silently; FAIL if accepted without a flag.	v_result (rule id)	Standard for the value (STA-021 3.2, 3.1 registries are normative); Proposed for the handling	S (executed: M10 DETECTED by R11 only; M4 by R5 only); E, L	Remove any flagged record
TC-04	2 / L1 and L2	As TC-01. For L2: PSAP B record from TC-01 exists.	TD-02R with incidentOwningAgencyIndicator true for both agencies (mutation M8).	L1: schema reports no error; rule R10 FAIL listing both agencies. L2: receiving system flags an ownership conflict for operator resolution and takes no automatic dispatch action.	L1: PASS if R10 FAIL names both; FAIL any other case. L2: PASS if the conflict is flagged and only one owner remains after resolution; FAIL if two owners persist or dispatch proceeds.	v_result (R10); operator prompt; record	Standard for the rule (STA-021 2.6: only one agency component may be true); Proposed for the handling	S (executed: M8 DETECTED by R10 only); E, L	Restore owner flags to the TD-02R state
TC-05	3 / L2	PSAP B service area boundary defined in the testbed; TD-02R record does not yet exist.	TD-02R with the CurrentIncident locationByValue changed to a civic address outside the PSAP B service area (address differs from the RoutingLocation).	Either responseToRequest 606 with no record created, or 200 with the record flagged for operator review and automatic dispatch withheld.	PASS if 606 with no record, or 200 with flag and no automatic dispatch; FAIL if accepted and dispatched without review.	code; record flag; t_display - t_recv	Standard for code 606 (STA-024 3.8.1); Proposed for the flag and dispatch hold	E, L (S: OBSERVE only, address change verified)	Delete record
TC-06	3 / L2	TC-01 L2 passed; PSAP B record shows status ResourcesEnroute with incidentComponent.lastUpdateTimeStamp T2.	Schema valid EIDO from PSAP A with incidentStatusCommonRegistryText Active and incidentComponent.lastUpdateTimeStamp T1 earlier than T2.	PSAP B keeps ResourcesEnroute; the message is logged and flagged as older than the stored record; the comparison uses the incident component time, not the top-level time.	PASS if status unchanged and the message is logged with a flag; FAIL if status reverts.	record; receive log; t_process	Standard for the principle (STA-021: an EIDO conveys current state); Proposed for the handling (neither standard defines ordering)	E, L	Restore record status
TC-07	3 / L2	TC-01 L2 passed; record exists.	The identical TD-02R frame delivered a second time with the same requestId; variant with a new requestId.	Second receipt logged (second EidoLogEvent incoming); one PSAP B record; a response is returned for each request.	PASS if record count is 1 and two receipts are logged; FAIL any other case.	record count; log count; code	Standard for logging (STA-024 3.14); Proposed for the single record (neither standard defines duplicate handling; requestId is the standard correlation	E, L	None

							means)		
TC-08	3 / L2	TC-01 L2 passed; operator at PSAP B has not spoken with PSAP A.	Operator opens the transferred record.	Operator confirms every item of the checklist in Section 4.5 from the record alone.	PASS if all items are confirmed; FAIL if any item requires a call to PSAP A.	checklist; d_use	Proposed	L (E: OBSERVE with a mock display)	None
TC-09	3 / L2	Clocks of PSAP A, PSAP B, and the Logger FE synchronized; logs accessible.	TC-01 executed once more with timing capture.	t_send, t_recv, t_ack, t_process, t_display recorded; d_ack and d_use computed; if no response, a 408 log entry exists at PSAP A.	OBSERVE in version 0.1 (no threshold). PASS or FAIL once a threshold is adopted and marked as validated.	all time symbols	Standard for the 408 log (STA-024 3.14); measurement definitions Proposed	E, L	None
TC-10	1 / L1 and L2	As TC-01.	TD-02R inflated with additional data until the serialized size exceeds 10 mebibytes; control: TD-02R unaltered (10,840 bytes).	L1: rule R12 FAIL above 10 MiB, PASS for the control. L2: receiver returns 513 only for the inflated frame and never for the control.	PASS if the control is accepted and the inflated frame receives 513 or is accepted (a recipient MAY support more); FAIL if the control receives 513.	R12; code	Standard. STA-024 3.8.1: recipient maximum MUST be at least ten mebibytes	S (R12 executed on the control: PASS); E, L	None
TC-11	3 / L2	Testbed receiver configured to return 403, 603, or 606 in turn.	TD-02R sent; receiver returns each configured code.	Sender takes the action in Section 3.4 for the code: no unchanged retry for 403; operator notified for 403, 603, 606; alternate agency sought for 603.	PASS if the sender's action matches Section 3.4 for every code; FAIL any other case.	sender log; operator notice	Standard for code meanings (STA-024 3.8.1); Proposed for the operator notice	E, L	Reset receiver configuration

4.4 Test descriptions

Each description follows the structure of the Texas A&M NG9-1-1 test description template: overview (summary, description, references), configuration (interfaces and whether the test system is active on them or only monitors), pre-test conditions, test sequence (preamble, numbered body steps, verdict), post-test conditions, and comments. Steps marked (S) are executed in the static environment, (E) in the emulated testbed, (L) on live systems; a step without a mark applies to E and L. Every verdict lists its PASS conditions and its FAIL conditions separately; a case is PASS only when every PASS condition is met.

TC-01. Initial transfer of an active incident

Suite and level: Suite 1, Level 1 and Level 2

Summary. A request to take over an active incident, carrying a valid EIDO, is accepted and becomes a record at PSAP B.

Description. PSAP A sends requestForService with fixture TD-02R. Level 1 checks that the EIDO itself is valid: it passes the published schema and the twelve document rules. Level 2 checks what NENA-STA-024 requires of a recipient that answers 200: an active incident is created at PSAP B from the EIDO, and the receipt is logged.

References.

- NENA-STA-021 sections 2.4, 2.6, 2.9, 2.18 (content); openapi.yaml required lists
- NENA-STA-024 section 3.8.1 (RequestForService; a recipient that responds 200 MUST create an active incident based on the EIDO)
- NENA-STA-024 section 3.14 (any FE that sends or receives an EIDO MUST log it with an EidoLogEvent)
- Element table rows: EIDO.$id, eidoVersion, issuingElementIdentification, incidentComponent (type, status, locationReference), agencyComponent (both agencies)

Configuration.

Party / interface	Connection	Mode
PSAP A emulator, interface IF_A_PUSH	connected to IF_B_PUSH	Active
PSAP B (IUT), interface IF_B_PUSH	connected to IF_A_PUSH	Active
PSAP B (IUT), interface IF_B_LOG	connected to Logger FE	Passive (monitored)
PSAP B record store and operator display	queried after the stimulus	Passive (observed)

Pre-test conditions.

Static environment (S):

- One computer with Python 3.9 or later, packages jsonschema and pyyaml installed.
- Files present in one folder: openapi.yaml (schema version 1.0.1), the fixture named in the case, eido_static_check_v1_1.py (Appendix D).
- SHA-256 of openapi.yaml equals 33d7261c3b7ae73facfca18268c2774455e1167d2f7c3c627ebb62b5a5789512.
- Fixture: TD-02R_request_payload.json, SHA-256 c04a620e092a66a59d481b9799d8b2358f10b02cc96a41c5809aa4b24a601e58.

Emulated testbed or live systems (E, L):

- PSAP A emulator and PSAP B implementation under test (IUT) reachable over TLS; certificates accepted on both sides.
- Push WebSocket session established from PSAP A to PSAP B (NENA-STA-024 section 3.8).
- A Logger FE, or a local log file standing in for it, receives EidoLogEvent records from PSAP B.
- Clocks of PSAP A, PSAP B, and the Logger FE synchronized to within one second (record the offset).
- IXIT sheet for PSAP B completed: incident type mapping table, service area boundary, operator display available, record query method.

- No record with Incident Tracking Identifier urn:emergency:uid:incidentid:a56e556d871:bcf.state.pa.us exists at PSAP B.

Test sequence.

Preamble:

- (S) Open a terminal or notebook in the fixture folder.
- (E, L) Start packet capture on IF_A_PUSH. Note the capture start time.
- (E, L) Generate a requestId (UUID) and record it on the result sheet.

Body:

Step	Action (stimulus)	Expected response	Observation recorded
1	(S) Run: python eido_static_check_v1_1.py --schema openapi.yaml --sample TD-02R_request_payload.json --operator "<name>" --out TC-01_L1.txt	Report written. Line 2 reads "Schema validation ... PASS, 0 errors". Rules R1 to R12 each show PASS (NA is allowed only for R7 and R11 on a fixture without an incident component; TD-02R has one, so none should be NA).	v_result: number of schema errors; list of rules not PASS
2	(E, L) At t_send, PSAP A sends a WebSocket text frame: {"requestForService": {"requestId": "<uuid>", "eido": <TD-02R>}}	Frame accepted by the transport (no close frame, no transport error).	t_send
3	(E, L) Observe the Logger FE.	An EidoLogEvent with direction incoming and requestId equal to the sent value appears, with the EIDO body.	t_recv; requestId match (Y/N)
4	(E, L) Observe IF_A_PUSH for the response.	A responseToRequest frame with requestId equal to the sent value and responseCode 200. A provisional 180 or 182 before the 200 is allowed.	t_ack; code
5	(E, L) Query the PSAP B record store for the Incident Tracking Identifier of TD-02R.	One record exists. Its incident type is the PSAP B mapping of MVA (from the IXIT table); status Active; the dispatch location equals the civic address of the CurrentIncident location (country US, A1 DE, A3 Wilmington, RD Cellular, HNO 42); the owning agency shown is state.pa.us; PSAP B is shown with role TransferredTo.	t_process; five field comparisons (Y/N each)
6	(E, L) Observe the PSAP B operator display.	The record is visible to the operator without manual retrieval.	t_display

Verdict:

PASS when all of the following hold:

- Level 1: step 1 shows 0 schema errors AND no rule reports FAIL.
- Level 2: steps 2 to 6 all met: code 200 with matching requestId; EidoLogEvent incoming with matching requestId; one record with all five fields equal; record visible on the display.

FAIL when any of the following holds:

- Level 1: any schema error, or any rule FAIL.
- Level 2: no response within the local timeout (record a 408 log entry at PSAP A); a final code other than 200; no EidoLogEvent; no record, more than one record, or any of the five fields different; record not visible without manual retrieval.

Other verdicts:

- BLOCKED: the Push session cannot be established (precondition failure), or the IXIT incident type mapping is missing.
- NA: not applicable; this case applies to every receiving system.

Post-test conditions.

- (E, L) Delete the record created at PSAP B, or mark it as test data per PSAP B's policy.
- (E, L) Stop packet capture; save the capture and the result sheet with the requestId.

Comments.

- Executed in environment S on 8 November 2026 by the author: Level 1 PASS (Appendix A.2). Level 2 not executed.
- The five record fields are the minimum a dispatcher needs; a receiving agency may add fields from its own data dictionary.

TC-02. Missing required element in the incident component

Suite and level: Suite 1, Level 1

Summary. An EIDO whose incident component lacks its required incident type is rejected by validation and does not become a record.

Description. The schema marks incidentTypeCommonRegistryText as required inside IncidentInformationType. Removing it must produce a schema error that names the element, and a receiving system must not create a record from the EIDO. In environments E and L the response code must not be 200.

References.

- NENA-STA-021 section 2.9 (incident component); openapi.yaml IncidentInformationType required list
- NENA-STA-024 section 3.13 (discrepancy categories EIDOMissingElement, NonConformant; reporting is MAY)
- Element table rows: incidentComponent.incidentTypeCommonRegistryText

Configuration.

Party / interface	Connection	Mode
Static environment	fixture folder	Active
PSAP A emulator and PSAP B (IUT)	as TC-01	Active / Passive as TC-01

Pre-test conditions.

Static environment (S):

- One computer with Python 3.9 or later, packages jsonschema and pyyaml installed.
- Files present in one folder: openapi.yaml (schema version 1.0.1), the fixture named in the case, eido_static_check_v1_1.py (Appendix D).
- SHA-256 of openapi.yaml equals 33d7261c3b7ae73facfca18268c2774455e1167d2f7c3c627ebb62b5a5789512.
- Fixture: TD-02R with the key incidentTypeCommonRegistryText deleted from incidentComponent (save as TD-02R_M_notype.json; record its SHA-256).

Emulated testbed or live systems (E, L):

- PSAP A emulator and PSAP B implementation under test (IUT) reachable over TLS; certificates accepted on both sides.
- Push WebSocket session established from PSAP A to PSAP B (NENA-STA-024 section 3.8).
- A Logger FE, or a local log file standing in for it, receives EidoLogEvent records from PSAP B.
- Clocks of PSAP A, PSAP B, and the Logger FE synchronized to within one second (record the offset).
- IXIT sheet for PSAP B completed: incident type mapping table, service area boundary, operator display available, record query method.
- No record with the TD-02R Incident Tracking Identifier exists at PSAP B.

Test sequence.

Preamble:

- (S) Create the mutated fixture with a text editor or a three-line script; confirm it is still valid JSON.
- (E, L) As TC-01.

Body:

Step	Action (stimulus)	Expected response	Observation recorded
1	(S) Run the check script on TD-02R_M_notype.json.	Line 2 reads "Schema validation ... FAIL, 1 errors" (or more) and the error path contains incidentComponent and the message names incidentTypeCommonRegistryText as a required property.	v_result: error text
2	(E, L) PSAP A sends requestForService with the mutated EIDO.	responseToRequest with a final code other than 200 (expected 488 Not Acceptable Here or a 5xx; the standard does not fix the code for a content defect). If PSAP B reports discrepancies, the category is EIDOMissingElement or NonConformant.	code; discrepancy report (if any)
3	(E, L) Query the PSAP B record store.	No record with the TD-02R Incident Tracking Identifier.	record count

Verdict:

PASS when all of the following hold:

- Step 1: at least one schema error whose path includes incidentComponent and whose message names incidentTypeCommonRegistryText.
- Steps 2 and 3 (E, L): final code other than 200 AND record count 0.

FAIL when any of the following holds:

- Step 1 reports 0 errors, or the error does not name the element.
- Code 200 returned for the mutated EIDO, or a record is created.

Other verdicts:

- OBSERVE: the specific non-200 code returned is recorded but not judged.
- BLOCKED: as TC-01.

Post-test conditions.

- (E, L) Confirm no record remains; stop capture.

Comments.

- Environment S result on 8 November 2026: mutations of the same kind (M1, M3, M9: removal of required properties at top level, agency, and call) were all reported by the schema (Appendix A). The exact mutation of this case (incident type removed) was not among the ten scripted mutations and should be added to the script in the next version.

TC-03. Registry value outside the registry

Suite and level: Suite 2, Level 1

Summary. A coded value that is not in its NENA registry passes the schema and must be caught by a rule check.

Description. The schema types registry fields as strings and does not list the registry values. An incident type or an agency role that is not in the registry is therefore schema valid. A receiving system that relies on the schema alone accepts it. The case shows that a rule check reports the value and that the receiving system does not create a record silently.

References.

- NENA-STA-021 section 3.2 (EIDO-IncidentType-Common registry, 245 codes, from APCO ANS 2.103.2-2019) and section 3.1 (EIDO-AgencyRole registry)
- NENA-STA-024 section 3.13 (discrepancy category BadDomain)
- Element table rows: incidentTypeCommonRegistryText, agencyRoleDescriptionRegistryText

Configuration.

Party / interface	Connection	Mode
Static environment	fixture folder	Active
PSAP A emulator and PSAP B (IUT)	as TC-01	Active / Passive as TC-01

Pre-test conditions.

Static environment (S):

- One computer with Python 3.9 or later, packages jsonschema and pyyaml installed.
- Files present in one folder: openapi.yaml (schema version 1.0.1), the fixture named in the case, eido_static_check_v1_1.py (Appendix D).
- SHA-256 of openapi.yaml equals 33d7261c3b7ae73facfca18268c2774455e1167d2f7c3c627ebb62b5a5789512.
- Fixture: TD-02R (the script creates the mutations M10 and M4 in memory).

Emulated testbed or live systems (E, L):

- PSAP A emulator and PSAP B implementation under test (IUT) reachable over TLS; certificates accepted on both sides.
- Push WebSocket session established from PSAP A to PSAP B (NENA-STA-024 section 3.8).
- A Logger FE, or a local log file standing in for it, receives EidoLogEvent records from PSAP B.
- Clocks of PSAP A, PSAP B, and the Logger FE synchronized to within one second (record the offset).
- IXIT sheet for PSAP B completed: incident type mapping table, service area boundary, operator display available, record query method.
- Mutated fixtures saved as files: TD-02R_M10.json (incident type INVALID_CODE) and TD-02R_M4.json (agency role Transferring).

Test sequence.

Preamble:

- (S) None beyond the common preconditions.
- (E, L) As TC-01.

Body:

Step	Action (stimulus)	Expected response	Observation recorded
1	(S) Run the check script on TD-02R.	In section 4 of the report, line M10 reads DETECTED [rules: R11] and line M4 reads DETECTED [rules: R5]; neither line shows a schema error count.	detection lines
2	(E, L) Send TD-02R_M10.json, then TD-02R_M4.json, each with a new requestId.	For each: either a final code other than 200 and no record, or code 200 with the record flagged for operator review and the offending value shown. If a discrepancy report is produced, its category is BadDomain.	code; record flag; discrepancy
3	(E, L) Query the record store and the operator display.	No unflagged record exists for either mutated EIDO.	record state

Verdict:

PASS when all of the following hold:

- Step 1: M10 detected by R11 only and M4 detected by R5 only (schema error count 0 on both).
- Steps 2 and 3 (E, L): for both mutated EIDOs, either no record, or a flagged record with the offending value visible.

FAIL when any of the following holds:

- Step 1: M10 or M4 reported as NOT DETECTED.
- A mutated EIDO becomes an unflagged record.

Other verdicts:

- OBSERVE: whether PSAP B rejects or accepts with a flag is recorded; both are PASS. The standard does not choose between them.

Post-test conditions.

- (E, L) Remove any flagged records.

Comments.

- Executed in environment S on 8 November 2026: M10 DETECTED [rules: R11], M4 DETECTED [rules: R5] (Appendix A.2, A.3).

TC-04. Two agencies marked as owning the incident

Suite and level: Suite 2, Level 1 and Level 2

Summary. An EIDO with two owning agencies passes the schema; the rule check and the receiving operator must both catch the conflict.

Description. NENA-STA-021 allows only one agency component to carry incidentOwningAgencyIndicator true. The schema does not enforce this. Level 1 shows the rule check reports both agencies. Level 2 shows that PSAP B does not act on a conflicting handover without an operator decision.

References.

- NENA-STA-021 section 2.6 (only one agency may be marked as owner; once set true, the agency that set it should clear it)
- Element table row: agencyComponent[].incidentOwningAgencyIndicator

Configuration.

Party / interface	Connection	Mode
Static environment	fixture folder	Active
PSAP A emulator and PSAP B (IUT)	as TC-01	Active / Passive as TC-01
PSAP B operator position	observed	Passive

Pre-test conditions.

Static environment (S):

- One computer with Python 3.9 or later, packages jsonschema and pyyaml installed.
- Files present in one folder: openapi.yaml (schema version 1.0.1), the fixture named in the case, eido_static_check_v1_1.py (Appendix D).
- SHA-256 of openapi.yaml equals 33d7261c3b7ae73facfca18268c2774455e1167d2f7c3c627ebb62b5a5789512.
- Fixture: TD-02R (the script creates mutation M8 in memory).

Emulated testbed or live systems (E, L):

- PSAP A emulator and PSAP B implementation under test (IUT) reachable over TLS; certificates accepted on both sides.
- Push WebSocket session established from PSAP A to PSAP B (NENA-STA-024 section 3.8).
- A Logger FE, or a local log file standing in for it, receives EidoLogEvent records from PSAP B.
- Clocks of PSAP A, PSAP B, and the Logger FE synchronized to within one second (record the offset).
- IXIT sheet for PSAP B completed: incident type mapping table, service area boundary, operator display available, record query method.
- TC-01 Level 2 has passed and its record exists at PSAP B (owner state.pa.us).
- Mutated fixture saved: TD-02R_M8.json (both agencies true).

Test sequence.

Preamble:

- (E, L) An operator is logged in at PSAP B and can be asked what the display shows.

Body:

Step	Action (stimulus)	Expected response	Observation recorded
1	(S) Run the check script on TD-02R.	Line M8 reads DETECTED [rules: R10] with no schema error count; on the unmutated fixture R10 shows one owner, state.pa.us.	detection line; owners list
2	(E, L) Send TD-02R_M8.json with a new requestId.	PSAP B responds (code recorded) and does not change the owner shown on the existing record automatically. An operator prompt or flag reports the conflict and names both agencies.	code; prompt text; owner field before and after
3	(E, L) Operator resolves the prompt by selecting one agency.	The record shows one owner; no dispatch action was triggered between step 2 and step 3.	owner field; dispatch log

Verdict:

PASS when all of the following hold:

- Level 1: M8 DETECTED by R10 only.
- Level 2: owner unchanged at step 2, conflict flagged naming both agencies, one owner after step 3, no dispatch action in between.

FAIL when any of the following holds:

- Level 1: M8 NOT DETECTED.
- Level 2: owner changed automatically, or two owners persist, or a dispatch action occurred before resolution.

Other verdicts:

- BLOCKED: no operator available for step 3 (record steps 1 and 2 only).
- OBSERVE: the response code at step 2 is recorded, not judged.

Post-test conditions.

- (E, L) Restore the record owner to state.pa.us or delete the record.

Comments.

- Executed in environment S on 8 November 2026: M8 DETECTED [rules: R10] (Appendix A). Level 2 not executed.

TC-05. Incident location outside the receiving agency's service area

Suite and level: Suite 3, Level 2

Summary. A transfer whose scene lies outside PSAP B's service area is either declined with 606 or accepted with an operator flag, never dispatched automatically.

Description. NENA-STA-024 defines code 606 for an agency that does not serve the requested area. The standard does not require a recipient to check the location before answering; this case proposes that it does, or that it flags the record. Automatic dispatch to a scene outside the service area is the failure to prevent.

References.

- NENA-STA-024 section 3.8.1 (code 606 Not Acceptable: the agency does not serve the area)
- NENA-STA-021 section 2.18 and registry 3.8 (location type CurrentIncident)
- Element table rows: locationComponent (CurrentIncident), incidentComponent.locationReference

Configuration.

Party / interface	Connection	Mode
PSAP A emulator and PSAP B (IUT)	as TC-01	Active / Passive as TC-01
PSAP B service area definition (IXIT)	configured before the test	Passive

Pre-test conditions.

Static environment (S):

- Not executable statically beyond confirming that the mutated address differs from the original (OBSERVE).

Emulated testbed or live systems (E, L):

- PSAP A emulator and PSAP B implementation under test (IUT) reachable over TLS; certificates accepted on both sides.
- Push WebSocket session established from PSAP A to PSAP B (NENA-STA-024 section 3.8).
- A Logger FE, or a local log file standing in for it, receives EidoLogEvent records from PSAP B.
- Clocks of PSAP A, PSAP B, and the Logger FE synchronized to within one second (record the offset).
- IXIT sheet for PSAP B completed: incident type mapping table, service area boundary, operator display available, record query method.
- Service area of PSAP B defined in the IXIT sheet as a civic boundary or polygon.
- Mutated fixture saved: TD-02R_M_outside.json, in which the CurrentIncident locationByValue civic address is replaced by an address outside the boundary (change A1, A3, RD, and HNO; keep the XML well formed).

Test sequence.

Preamble:

- Confirm with the PSAP B operator that the mutated address is outside the service area before sending.

Body:

Step	Action (stimulus)	Expected response	Observation recorded
1	Send TD-02R_M_outside.json with a new requestId.	Either (a) responseToRequest 606 and no record, or (b) responseToRequest 200 and a record flagged "outside service area" (or equivalent) on the operator display.	code; record; flag
2	Observe the dispatch log for 60 seconds after the response (proposed observation window).	No unit assignment or dispatch action on the record without an operator action.	dispatch log
3	Send the unmutated TD-02R as a control with a new requestId.	Code 200 and an unflagged record (as TC-01).	code; record

Verdict:

PASS when all of the following hold:

- Step 1 outcome (a) or (b), AND step 2 shows no automatic dispatch, AND step 3 control passes.

FAIL when any of the following holds:

- Step 1: code 200 with an unflagged record for the outside address; or step 2: a dispatch action without an operator; or step 3 control fails.

Other verdicts:

- OBSERVE (S): only that the address was changed and the XML remains well formed.
- BLOCKED: no service area defined in the IXIT sheet.

Post-test conditions.

- Delete both records (mutated and control).

Comments.

- Proposed criterion. The 60 second window is a placeholder to be replaced after consultation with practitioners.

TC-06. Older update must not overwrite a newer record

Suite and level: Suite 3, Level 2

Summary. An EIDO whose incident component is older than the record at PSAP B is logged and flagged but does not change the record.

Description. The EIDO conveys the current state of an incident. Neither standard defines ordering of updates arriving out of sequence. This case proposes that PSAP B compares the incident component's lastUpdateTimeStamp with its record and keeps the newer state.

References.

- NENA-STA-021 section 1 and 2.9 (an EIDO conveys the current state; incidentComponent.lastUpdateTimeStamp)
- Element table rows: incidentComponent.lastUpdateTimeStamp, incidentStatusCommonRegistryText

Configuration.

Party / interface	Connection	Mode
PSAP A emulator and PSAP B (IUT)	as TC-01	Active / Passive as TC-01

Pre-test conditions.

Static environment (S):

- Not executable statically.

Emulated testbed or live systems (E, L):

- PSAP A emulator and PSAP B implementation under test (IUT) reachable over TLS; certificates accepted on both sides.
- Push WebSocket session established from PSAP A to PSAP B (NENA-STA-024 section 3.8).
- A Logger FE, or a local log file standing in for it, receives EidoLogEvent records from PSAP B.
- Clocks of PSAP A, PSAP B, and the Logger FE synchronized to within one second (record the offset).
- IXIT sheet for PSAP B completed: incident type mapping table, service area boundary, operator display available, record query method.
- TC-01 Level 2 passed; PSAP B record exists with status Active.
- A second fixture TD-02R_T2.json: status ResourcesEnroute, incidentComponent.lastUpdateTimeStamp T2 = 2021-04-30T14:50:00.000-04:00, top-level timestamp T2 or later.
- A third fixture TD-02R_T1.json: status Active, incidentComponent.lastUpdateTimeStamp T1 = 2021-04-30T14:44:00.000-04:00 (older than T2), top-level timestamp later than T2 (to show the comparison must use the component time).

Test sequence.

Preamble:

- Record the PSAP B record's status and time fields before step 1.

Body:

Step	Action (stimulus)	Expected response	Observation recorded
1	Send TD-02R_T2.json as an update of the same incident (same Incident Tracking Identifier, new requestId).	Record status becomes ResourcesEnroute with time T2.	record status; time
2	Send TD-02R_T1.json (older component time, newer top-level time) with a new requestId.	Record status remains ResourcesEnroute with time T2; the receipt is logged (EidoLogEvent incoming) and flagged as older than the stored record.	record status; log entry; flag
3	Inspect the flag.	The flag or log shows the two times compared (T1 and T2) and names the component.	flag content

Verdict:

PASS when all of the following hold:

- Step 1 updates the record; step 2 leaves it unchanged with a logged, flagged receipt; step 3 shows the component times were compared.

FAIL when any of the following holds:

- Status reverts to Active at step 2, or the receipt is not logged, or the comparison used the top-level time (record changed because the top-level time was newer).

Other verdicts:

- BLOCKED: PSAP B does not accept updates to an existing record over the Push transport (record the mechanism it uses instead).

Post-test conditions.

- Restore the record to its pre-test state or delete it.

Comments.

- Proposed criterion. Whether updates to a transferred incident arrive by Push or by subscription (NENA-STA-024 section 3.3) depends on the deployment; the case is written for Push and is to be adapted if subscription is used.

TC-07. Duplicate delivery of the same request

Suite and level: Suite 3, Level 2

Summary. The same requestForService delivered twice produces two log entries, two responses, and one record.

Description. Transport retries can deliver a frame twice. The requestId that the responder returns is the standard means of relating a response to a request. The case proposes that a second delivery with the same requestId, or the same EIDO with a new requestId, does not create a second record.

References.

- NENA-STA-024 section 3.8.1 (requestId generated by the requester and returned by the responder), section 3.14 (every received EIDO is logged)
- Element table row: EIDO.$id (Incident Tracking Identifier)

Configuration.

Party / interface	Connection	Mode
PSAP A emulator and PSAP B (IUT)	as TC-01	Active / Passive as TC-01
Logger FE	observed	Passive

Pre-test conditions.

Static environment (S):

- Not executable statically.

Emulated testbed or live systems (E, L):

- PSAP A emulator and PSAP B implementation under test (IUT) reachable over TLS; certificates accepted on both sides.
- Push WebSocket session established from PSAP A to PSAP B (NENA-STA-024 section 3.8).
- A Logger FE, or a local log file standing in for it, receives EidoLogEvent records from PSAP B.
- Clocks of PSAP A, PSAP B, and the Logger FE synchronized to within one second (record the offset).
- IXIT sheet for PSAP B completed: incident type mapping table, service area boundary, operator display available, record query method.
- No record with the TD-02R Incident Tracking Identifier exists at PSAP B.

Test sequence.

Preamble:

- Record the Logger FE entry count for PSAP B before step 1.

Body:

Step	Action (stimulus)	Expected response	Observation recorded
1	Send TD-02R with requestId Q1.	Code 200; one record; one EidoLogEvent incoming with Q1.	code; record count; log count
2	Send the byte-identical frame again (same Q1).	A response is returned (code recorded); the Logger FE shows a second EidoLogEvent incoming with Q1; the record count for the Incident Tracking Identifier is still 1.	code; log count; record count
3	Send TD-02R again with a new requestId Q2 (same EIDO content).	A response is returned; a third log entry; record count still 1.	code; log count; record count

Verdict:

PASS when all of the following hold:

- Record count 1 after steps 2 and 3, AND three EidoLogEvent entries, AND a response to each request.

FAIL when any of the following holds:

- Record count greater than 1 at any step, or a receipt not logged, or a request left without a response.

Other verdicts:

- OBSERVE: the response codes at steps 2 and 3 (200 again, or another code) are recorded; the standard does not define them.

Post-test conditions.

- Delete the record; save the three log entries.

Comments.

- Proposed criterion; the standards define the logging and the requestId but not the duplicate handling.

TC-08. Operator can act on the transferred record without calling PSAP A

Suite and level: Suite 3, Level 2

Summary. A PSAP B operator confirms the seven checklist items from the record alone.

Description. The purpose of a transfer is that the receiving operator can act. The case uses the checklist of Section 4.5: each item must be answerable from the record, without a telephone call to PSAP A.

References.

- Element table rows: incident type, CurrentIncident location, status and time, owning agency, agency roles, internalIncidentId, callStartTimestamp
- Section 4.5 checklist

Configuration.

Party / interface	Connection	Mode
PSAP B operator position	observed by the tester	Passive

Pre-test conditions.

Static environment (S):

- Not executable statically.

Emulated testbed or live systems (E, L):

- TC-01 Level 2 passed; the record is on the operator display.
- The operator has not spoken with PSAP A about this incident.

- A tester with the checklist and a stopwatch is present (or a screen recording with timestamps is available).

Test sequence.

Preamble:

- Explain to the operator that the test is of the system's presentation, not of the operator.

Body:

Step	Action (stimulus)	Expected response	Observation recorded
1	Ask the operator to open the transferred record and start the timer.	Record opens.	t_display (if not already recorded)
2	Read each checklist item aloud and ask the operator to point to the answer on the screen.	For each of the seven items the operator points to a field and states the value; the tester compares with TD-02R.	seven Y/N marks; wrong values noted
3	Stop the timer after the seventh item.	Elapsed time recorded.	elapsed seconds

Verdict:

PASS when all of the following hold:

- All seven items marked Y with values matching TD-02R.

FAIL when any of the following holds:

- Any item marked N (not on the screen, or the operator would need to call PSAP A), or any value different from TD-02R.

Other verdicts:

- OBSERVE (E): with a mock display, record which items the mock can show; no verdict.
- Elapsed time is recorded without a threshold in this version.

Post-test conditions.

- None.

Comments.

- Proposed criterion. The checklist is the author's; a receiving agency may add items.

TC-09. Timing of delivery, acknowledgment, and display

Suite and level: Suite 3, Level 2 (OBSERVE in this version)

Summary. The five time points of Section 4.2 are captured for one transfer and the two intervals computed; no threshold is applied yet.

Description. Repeats TC-01 with timing capture. The purpose in this version is to establish how the times are read in a given environment, so that a later version can propose thresholds from measured values.

References.

- NENA-STA-024 section 3.14 (when no response is received, the sender logs with status 408)
- Section 4.2 measurement definitions

Configuration.

Party / interface	Connection	Mode
PSAP A emulator, PSAP B (IUT), Logger FE	as TC-01	Active / Passive as TC-01
Clock reference	NTP or GPS source stated in the result	Passive

Pre-test conditions.

Static environment (S):

- Not executable statically.

Emulated testbed or live systems (E, L):

- PSAP A emulator and PSAP B implementation under test (IUT) reachable over TLS; certificates accepted on both sides.
- Push WebSocket session established from PSAP A to PSAP B (NENA-STA-024 section 3.8).
- A Logger FE, or a local log file standing in for it, receives EidoLogEvent records from PSAP B.
- Clocks of PSAP A, PSAP B, and the Logger FE synchronized to within one second (record the offset).
- IXIT sheet for PSAP B completed: incident type mapping table, service area boundary, operator display available, record query method.
- Clock offsets between the three systems measured and written on the result sheet.

Test sequence.

Preamble:

- Enable timestamped logging at PSAP A (send and receive), the Logger FE, PSAP B record store, and the operator display (screen recording with a visible clock is acceptable).

Body:

Step	Action (stimulus)	Expected response	Observation recorded
1	Execute TC-01 steps 2 to 6.	TC-01 Level 2 PASS.	all TC-01 observations
2	Read t_send, t_recv, t_ack, t_process, t_display from the logs.	Five timestamps with source named for each.	five times
3	Compute d_ack = t_ack - t_send and d_use = t_display - t_send, corrected by the measured clock offsets.	Two intervals in seconds with the correction shown.	d_ack; d_use
4	Variant: configure PSAP B not to respond; send TD-02R; wait for the PSAP A local timeout.	PSAP A logs the request with status 408.	408 log entry; timeout value

Verdict:

PASS when all of the following hold:

- Not applicable in this version (OBSERVE). Step 4: PASS if the 408 log entry exists; FAIL if PSAP A records nothing or records 200.

FAIL when any of the following holds:

- Step 4 only: no 408 entry.

Other verdicts:

- OBSERVE: d_ack and d_use are recorded. A later version will state thresholds marked "proposed threshold, subject to validation".

Post-test conditions.

- Restore PSAP B response behavior; delete the record.

Comments.

- Measurement definitions are the author's; the 408 logging is a standard requirement.

TC-10. Message size limit

Suite and level: Suite 1, Level 1 and Level 2

Summary. An EIDO at normal size is never refused as too large; one above 10 mebibytes may be refused with 513.

Description. NENA-STA-024 requires a recipient to accept messages of at least ten mebibytes and defines code 513 for messages above its maximum. Level 1 measures the serialized size of the fixture (rule R12). Level 2 sends a control and an inflated EIDO.

References.

- NENA-STA-024 section 3.8.1 (code 513 Message Too Large; recipient maximum MUST be at least 10 MiB)
- Rule R12 (Appendix D)

Configuration.

Party / interface	Connection	Mode
Static environment	fixture folder	Active
PSAP A emulator and PSAP B (IUT)	as TC-01	Active / Passive as TC-01

Pre-test conditions.

Static environment (S):

- One computer with Python 3.9 or later, packages jsonschema and pyyaml installed.
- Files present in one folder: openapi.yaml (schema version 1.0.1), the fixture named in the case, eido_static_check_v1_1.py (Appendix D).
- SHA-256 of openapi.yaml equals 33d7261c3b7ae73facfca18268c2774455e1167d2f7c3c627ebb62b5a5789512.
- Fixture: TD-02R (10,840 bytes serialized).

Emulated testbed or live systems (E, L):

- PSAP A emulator and PSAP B implementation under test (IUT) reachable over TLS; certificates accepted on both sides.
- Push WebSocket session established from PSAP A to PSAP B (NENA-STA-024 section 3.8).
- A Logger FE, or a local log file standing in for it, receives EidoLogEvent records from PSAP B.
- Clocks of PSAP A, PSAP B, and the Logger FE synchronized to within one second (record the offset).
- IXIT sheet for PSAP B completed: incident type mapping table, service area boundary, operator display available, record query method.
- Inflated fixture TD-02R_big.json: TD-02R with additional data components added until the serialized size exceeds 10,485,760 bytes (for example, a long additionalDataByValue XML comment string); must remain schema valid.

Test sequence.

Preamble:

- (S) None.
- (E, L) Confirm the inflated file size with the operating system and record it.

Body:

Step	Action (stimulus)	Expected response	Observation recorded
1	(S) Run the check script on TD-02R.	R12 PASS with the byte count (10,840 for the unaltered fixture).	R12 line
2	(S) Run the check script on TD-02R_big.json.	R12 FAIL with a byte count above 10,485,760; schema still PASS.	R12 line; schema line
3	(E, L) Send TD-02R (control) with a new requestId.	Code 200 (as TC-01).	code
4	(E, L) Send TD-02R_big.json with a new requestId.	Either code 513, or code 200 (a recipient may support more than 10 MiB). The transport does not close the session abnormally.	code; session state

Verdict:

PASS when all of the following hold:

- Steps 1 and 2 as expected; step 3 code 200; step 4 code 513 or 200 with the session intact.

FAIL when any of the following holds:

- Step 3 returns 513 (control refused), or step 4 closes the session without a response, or R12 misreports.

Other verdicts:

- OBSERVE: PSAP B's actual maximum, if stated in its IXIT sheet, is recorded.

Post-test conditions.

- Delete any records; remove the inflated file from the fixture folder.

Comments.

- Executed in environment S on 8 November 2026 for the control only: R12 PASS, 10,840 bytes (Appendix A.2). The inflated fixture has not been built.

TC-11. Sender handling of refusal codes

Suite and level: Suite 3, Level 2

Summary. When PSAP B refuses with 403, 603, or 606, PSAP A takes the action the response table gives and informs its operator.

Description. The response table of NENA-STA-024 gives the sender's action for each code. This case tests the sending side: no unchanged retry after 403, an alternate agency sought after 603, and an operator notice for all three.

References.

- NENA-STA-024 section 3.8.1 (ResponseToRequest status table and notes)
- Section 3.4 of this document

Configuration.

Party / interface	Connection	Mode
PSAP A (IUT in this case) and PSAP B emulator	roles reversed from TC-01	PSAP A Active; PSAP B emulator Active
PSAP A operator position	observed	Passive

Pre-test conditions.

Static environment (S):

- Not executable statically.

Emulated testbed or live systems (E, L):

- Push session established; PSAP B emulator configurable to return a chosen code.
- PSAP A has a list of alternate agencies for the incident type (IXIT).

Test sequence.

Preamble:

- Configure the PSAP B emulator to return 403 with responseText "policy".

Body:

Step	Action (stimulus)	Expected response	Observation recorded
1	PSAP A sends TD-02R.	PSAP A receives 403; within the observation window it does not resend the same request unchanged; the operator sees a notice naming the code and the agency.	send log; operator notice
2	Reconfigure the emulator to return 603; PSAP A sends TD-02R.	PSAP A receives 603; the send log shows a request to an alternate agency, or an operator prompt offering one; operator notice shown.	send log; notice
3	Reconfigure the emulator to return 606; PSAP A sends TD-02R.	PSAP A receives 606; operator notice shown with the code meaning (area not served); no automatic resend to the same agency.	send log; notice

Verdict:

PASS when all of the following hold:

- All three steps: correct code received, action matches Section 3.4, operator notice present.

FAIL when any of the following holds:

- An unchanged resend to the same agency after 403 or 606, or no operator notice, or no alternate sought after 603.

Other verdicts:

- OBSERVE: the wording of the operator notice is recorded, not judged.
- BLOCKED: PSAP A has no alternate agency list.

Post-test conditions.

- Reset the emulator; clear notices.

Comments.

- The code meanings and sender actions are standard; the operator notice is proposed.

4.5 Operator checklist for TC-08

Item	Source element	Confirmed from the record alone (Y/N)
Incident type in PSAP B's own terms	incidentTypeCommonRegistryText mapped to PSAP B's internal type	
Dispatchable incident location	locationComponent of type CurrentIncident (or InitialIncident) referenced from incidentComponent.locationReference	
Current status and its time	incidentStatusCommonRegistryText; incidentComponent.lastUpdateTimeStamp	
Which agency owns the incident now	incidentOwningAgencyIndicator	
Which agency is asking PSAP B to act, and in what role	agencyRoleDescriptionRegistryText of each agency	
PSAP A's reference number for follow-up	internalIncidentId	
Time the original call was received	callComponent[].callStartTimestamp	

4.6 Execution environments

Each case states the environments in which it can be executed. The table below is the plan for closing the gap between what has been executed and what the specification requires.

Environment	What it is	Cases executable	Status	Who
S. Static	One computer; the published schema; fixtures; the check script (Appendix D)	TC-01 L1, TC-02, TC-03, TC-04 L1, TC-10 L1; TC-05 as OBSERVE	Executed 8 November 2026 (Appendix A)	Author
E. Emulated testbed	Sender and receiver emulators implementing requestForService and responseToRequest over WebSocket with TLS; a minimal Logger FE; configurable receiver behavior	All cases except TC-08 (OBSERVE with a mock display)	Not built; next step	Author; open source EIDO libraries to be evaluated
L. Live systems	CAD systems of two agencies or a vendor laboratory	All cases	Requires agency or vendor agreement; IXIT (implementation information for testing) from the receiving system	Agencies, vendors; author as test designer

5. Results to Date

Static checks were executed by the author on a Windows 11 computer (Python 3.12.4, jsonschema 4.19.2, pyyaml 6.0.1) with script version 1.1 on 8 November 2026. The full reports are in Appendix A; the input file hashes in each report match the fixture hashes in Section 3.5 and the schema in the NENA911/EIDO-JSON repository.

Fixture	Run (local time)	Schema	Rules	Mutations	Reading
TD-01	18:17:38	PASS, 0 errors	R1 to R6, R8, R9, R12 PASS; R7, R11 NA (no incident component); R10 PASS, no owner	M1 to M9 all DETECTED	The published sample is valid but does not meet the preconditions of an incident transfer: no incident component, no owning agency, one agency only.
TD-02R	19:33:33	PASS, 0 errors	R1 to R12 all PASS; R10 owner state.pa.us; R11 MVA; R12 10,840 bytes	M1 to M10 all DETECTED	The request payload is schema valid and meets every document rule checked. Six defect classes (M4, M5, M6, M7, M8, M10) are not reported by the schema and are detected only by the rule checks.
TD-02A	18:17:38	PASS, 0 errors	R1 to R12 all PASS; R10 owner psap-b.example.org; R11 MVA; R12 10,847 bytes	M1 to M10 all DETECTED	The post-acceptance state is schema valid and meets every document rule checked.

Two conclusions follow for the specification. First, schema validation alone is not a sufficient Level 1 check: of ten deliberately introduced defects, the schema reports four (missing required properties and a version enumeration) and does not report six (values outside three registries, a malformed timestamp, both location forms present, a dangling reference, two owning agencies). A receiving agency that validates against the schema only would accept all six. Second, the published sample cannot serve as the input of a transfer test; a constructed fixture with an incident component and two agencies is required, which is why TD-02R and TD-02A exist.

No Level 2 case has been executed. The results table for Suite 3 is empty by design and is to be filled in the emulated testbed or with agencies.

ID	Executed	Date	Environment	Result
TC-01 L1	Y	2026-11-08	S	PASS (TD-02R)
TC-01 L2	N			Not executed
TC-02	Y (schema part)	2026-11-08	S	DETECTED (M1, M3, M9 analogues)
TC-03	Y (rule part)	2026-11-08	S	DETECTED (M10 by R11; M4 by R5)
TC-04 L1	Y	2026-11-08	S	DETECTED (M8 by R10)
TC-04 L2	N			Not executed
TC-05	N			Not executed
TC-06	N			Not executed
TC-07	N			Not executed
TC-08	N			Not executed
TC-09	N			Not executed
TC-10 L1	Y (control)	2026-11-08	S	PASS (R12, 10,840 bytes)
TC-11	N			Not executed

6. Relevance to Program Accountability

Interoperability investments by state and local agencies are reported to funding programs through work plans and periodic performance reports; the Emergency Management Performance Grant is one such program. A receiving-agency acceptance record of the kind defined in Part B, with the fixture hashes, the environment, the verdicts, and the dates, is evidence that an exchange investment produced usable incident data at the receiving end, rather than evidence that two systems exchanged bytes. This document does not assume that an Emergency Operations Center has the authority or an interface to receive transferred incident data; that question requires a separate analysis of EOC authorities and interfaces and is outside this document.

7. Limitations

- All fixtures are synthetic. TD-02R and TD-02A were constructed by the author from the published sample; the CurrentIncident location reuses the sample's address and is not a verified coordinate.
- The only environment executed is a single computer. Transport, TLS, credentials (NENA-STA-024 section 3.15), and the receiving system's behavior were not exercised. The static results are not an execution of any Level 2 case.
- TD-02A keeps issuingElementIdentification and the top-level updatedByAgencyReference of PSAP A while marking PSAP B as owner. It represents the content PSAP B would hold after acceptance; re-issuance by PSAP B, with its own issuer identification, is outside this version.
- Neither standard defines duplicate handling or update ordering. TC-06 and TC-07 are therefore Proposed criteria. The requestId of NENA-STA-024 is the correlation means available.
- The ownership handover order (which agency clears its indicator and when) is not defined by NENA-STA-021 section 2.6 beyond the rule that the agency that set true should clear it. Section 3.2 records an assumption.
- NENA-STA-024 section 2.3 states that where a REST/JSON interface is described, the yaml file in the NENA GitHub repository is normative. The Push transport is a WebSocket interface and this document cites the text tables; the only repository hyperlink found in the text is for the Logging Service.
- Measurements have no thresholds in this version.
- This document is not a certification procedure. Execution of its cases by an agency or vendor produces a test report, not a certificate or a declaration of conformity. Execution against a live receiving system requires implementation information (IXIT) from that system's operator.
- NENA-STA-021 is under revision (021.1.1-202Y draft in public review in 2026). Element names and rules are taken from NENA-STA-021.1a-2022 [1], the corrected edition noted in [2], and the repository schema version 1.0.1, and will be rechecked against the published revision.
- The document has not been externally reviewed.

8. Dissemination and Review Plan

This document is written for the agencies and bodies that specify, procure, or oversee incident data exchange between jurisdictions, and for the groups that maintain the EIDO and conveyance standards. Version 0.1 is being offered for review to a researcher in government information systems interoperability, to a university public safety communications group, and to the operators of the Texas A&M NG9-1-1 testing program. Comments received will be recorded in the document control table of the next version.

The schema versus document rule findings of Section 5 relate to an open issue in the NENA911/EIDO-JSON repository (issue 35 [9], opened 15 May 2026, concerning fields whose restrictions in the document are not represented in the schema). The author intends to add the reproducible static check results of Appendix A to that discussion rather than open a parallel issue, after confirming the scope of the existing issue.

9. References

[1] National Emergency Number Association. NENA Standard for Emergency Incident Data Object (EIDO), NENA-STA-021.1a-2022, approved 19 April 2022 (original NENA-STA-021.1-2021, ANSI approved 19 October 2021). Text, JSON schema (openapi.yaml, schema version 1.0.1), and sample instances also published at https://github.com/NENA911/EIDO-JSON, commit 1ea5750 of 5 June 2026.

[2] National Emergency Number Association. Document Publishing Error Notice, NENA Standard for Emergency Incident Data Object (EIDO), 8 January 2025 (identifies the corrected edition as NENA-STA-021.1b). https://www.nena.org/news/690644/

[3] National Emergency Number Association. NENA Standard for the Conveyance of Emergency Incident Data Objects (EIDOs) between Next Generation (NG9-1-1) Systems and Applications, NENA-STA-024.1.1-2025, ANSI approved 30 January 2025, published 7 February 2025.

[4] APCO International. APCO ANS 2.103.2-2019, Public Safety Communications Common Incident Types for Data Exchange (basis of the EIDO-IncidentType-Common registry as cited in NENA-STA-021 section 3.2).

[5] Klyne, G. and C. Newman. Date and Time on the Internet: Timestamps, RFC 3339, July 2002.

[6] Texas A&M University, Internet2 Technology Evaluation Center / CACN. NG9-1-1 Interoperability Testing Program, repository tamu-edu/ng911-dev, v5.6 release, commit 73ba4f2 of 12 October 2026, MIT License. https://github.com/tamu-edu/ng911-dev

[7] Federal Emergency Management Agency, IPAWS Program Management Office. IPAWS Guide for Independent Testing of Emergency Alert System Equipment, June 2012. https://www.fema.gov/sites/default/files/documents/fema_ipaws-guide-independent-testing-emergency-alert-system-equipment.pdf

[8] Institute for Defense Analyses. Public Safety and Emergency Management Communications Information Models Gap Analysis, IDA Document D-8728, 2017.

[9] NENA911/EIDO-JSON repository, issue 35, "(V2) Some fields are missing restriction defined in the document", opened 15 May 2026. https://github.com/NENA911/EIDO-JSON/issues/35

Appendix A. Static Check Reports

Reports as written by the check script, unedited. Operator line shows the name given at the command line and the operating system login and host name of the computer on which the script ran.

A.1 TD-01

```
EIDO static check report (script v1.1)
Run at: 2026-11-08T18:17:38-05:00
Operator: Jungeun Kim (OS login: janek; host: Jane)
Python 3.12.4 on Windows-11-10.0.26300-SP0; jsonschema 4.19.2; pyyaml 6.0.1


Input files (SHA-256):
  schema   33d7261c3b7ae73facfca18268c2774455e1167d2f7c3c627ebb62b5a5789512  openapi.yaml
  sample   4f98a553e6e095e25c96a377a068c18ea87fe1674c65199f9a2f7afe0d5bf57d  Sample call transfer EIDO.json
  standard ad84ab1511f44204c9921f438a80503c3318f84a9b50edb0d322ab736be4d3ba  NENA-STA-021.1b EIDO JSON_2022.docx


1. JSON parse: PASS
2. Schema validation (EmergencyIncidentDataObjectType): PASS, 0 errors
3. Rule checks on the sample:
   R1 $id uniqueness                                  PASS  11 ids, duplicates: none
   R2 $ref resolution                                 PASS  25 refs, unresolved: none
   R3 embedded XML well-formed                        PASS  4 XML strings, failures: none
   R4 RFC 3339 timestamps                             PASS  13 timestamps, malformed: none
   R5 EIDO-AgencyRole registry                        PASS  agencies with values outside registry: none
   R6 EIDO-LocationType registry                      PASS  locations with values outside registry: none
   R7 EIDO-IncidentStatus-Common registry             NA    no incidentComponent in this EIDO
   R8 location by value XOR by reference (2.18)       PASS  violations: none
   R9 additional data by value XOR by reference (2.19)  PASS  violations: none
   R10 at most one owning agency (2.6)                PASS  owners: none
   R11 EIDO-IncidentType-Common registry (3.2)        NA    no incidentComponent or no incident type in this EIDO
   R12 size vs 10 MiB recipient minimum (STA-024 3.8.1) PASS  8414 bytes serialized
4. Mutation tests (each is a broken copy of the sample; a defect is 'detected' if the schema or a rule reports it):
   M1 remove required issuingElementIdentification    DETECTED  [schema: 1 error(s)]  expected: schema
   M2 eidoVersion 2.0                                 DETECTED  [schema: 1 error(s)]  expected: schema (enum)
   M3 remove required agencyType                      DETECTED  [schema: 1 error(s)]  expected: schema
   M4 agency role outside registry                    DETECTED  [rules: R5]  expected: rule R5 (schema does not check registry
values)
   M5 malformed top-level timestamp                   DETECTED  [rules: R4]  expected: rule R4 (schema format: date-time is not
enforced by default)
   M6 location with both by-value and by-reference    DETECTED  [rules: R8]  expected: rule R8 (schema does not enforce XOR)
   M7 dangling $ref                                   DETECTED  [rules: R2]  expected: rule R2
   M8 two owning agencies                             DETECTED  [rules: R10]  expected: rule R10 (schema does not enforce single
owner)
   M9 remove required callStateRegistryText           DETECTED  [schema: 1 error(s)]  expected: schema

Scope statement: this is a static check of published files on a single computer. No live system, network transport, authentication,
or agency data was involved. It is not an execution of the test cases in the specification.
```

A.2 TD-02R

```
EIDO static check report (script v1.1)
Run at: 2026-11-08T19:33:33-05:00
Operator: Jungeun Kim (OS login: janek; host: Jane)
Python 3.12.4 on Windows-11-10.0.26300-SP0; jsonschema 4.19.2; pyyaml 6.0.1


Input files (SHA-256):
  schema   33d7261c3b7ae73facfca18268c2774455e1167d2f7c3c627ebb62b5a5789512  openapi.yaml
  sample   c04a620e092a66a59d481b9799d8b2358f10b02cc96a41c5809aa4b24a601e58  TD-02R_request_payload.json
  standard ad84ab1511f44204c9921f438a80503c3318f84a9b50edb0d322ab736be4d3ba  NENA-STA-021.1b EIDO JSON_2022.docx


1. JSON parse: PASS
2. Schema validation (EmergencyIncidentDataObjectType): PASS, 0 errors
3. Rule checks on the sample:
   R1 $id uniqueness                                  PASS  14 ids, duplicates: none
   R2 $ref resolution                                 PASS  29 refs, unresolved: none
```

```
   R3 embedded XML well-formed                          PASS  5 XML strings, failures: none
   R4 RFC 3339 timestamps                               PASS  16 timestamps, malformed: none
   R5 EIDO-AgencyRole registry                          PASS  agencies with values outside registry: none
   R6 EIDO-LocationType registry                        PASS  locations with values outside registry: none
   R7 EIDO-IncidentStatus-Common registry               PASS  values outside registry: none
   R8 location by value XOR by reference (2.18)         PASS  violations: none
   R9 additional data by value XOR by reference (2.19)  PASS  violations: none
   R10 at most one owning agency (2.6)                  PASS  owners: ['state.pa.us']
   R11 EIDO-IncidentType-Common registry (3.2)          PASS  value: 'MVA'
   R12 size vs 10 MiB recipient minimum (STA-024 3.8.1) PASS  10840 bytes serialized
4. Mutation tests (each is a broken copy of the sample; a defect is 'detected' if the schema or a rule reports it):
   M1 remove required issuingElementIdentification      DETECTED  [schema: 1 error(s)]  expected: schema
   M2 eidoVersion 2.0                                   DETECTED  [schema: 1 error(s)]  expected: schema (enum)
   M3 remove required agencyType                        DETECTED  [schema: 1 error(s)]  expected: schema
   M4 agency role outside registry                      DETECTED  [rules: R5]  expected: rule R5 (schema does not check registry
values)
   M5 malformed top-level timestamp                     DETECTED  [rules: R4]  expected: rule R4 (schema format: date-time is not
enforced by default)
   M6 location with both by-value and by-reference      DETECTED  [rules: R8]  expected: rule R8 (schema does not enforce XOR)
   M7 dangling $ref                                     DETECTED  [rules: R2]  expected: rule R2
   M8 two owning agencies                               DETECTED  [rules: R10]  expected: rule R10 (schema does not enforce single
owner)
   M9 remove required callStateRegistryText             DETECTED  [schema: 1 error(s)]  expected: schema
   M10 incident type outside registry                   DETECTED  [rules: R11]  expected: rule R11 (schema does not check registry
values)

Scope statement: this is a static check of published files on a single computer. No live system, network transport, authentication,
or agency data was involved. It is not an execution of the test cases in the specification.
```

A.3 TD-02A

```
EIDO static check report (script v1.1)
Run at: 2026-11-08T18:17:38-05:00
Operator: Jungeun Kim (OS login: janek; host: Jane)
Python 3.12.4 on Windows-11-10.0.26300-SP0; jsonschema 4.19.2; pyyaml 6.0.1


Input files (SHA-256):
  schema   33d7261c3b7ae73facfca18268c2774455e1167d2f7c3c627ebb62b5a5789512  openapi.yaml
  sample   c49537b16b1e650d14b3fde55c200b36bc56f22e9fcc01d114540b79a7898ca2  TD-02_author_constructed_incident_transfer.json
  standard ad84ab1511f44204c9921f438a80503c3318f84a9b50edb0d322ab736be4d3ba  NENA-STA-021.1b EIDO JSON_2022.docx


1. JSON parse: PASS
2. Schema validation (EmergencyIncidentDataObjectType): PASS, 0 errors
3. Rule checks on the sample:
   R1 $id uniqueness                                    PASS  14 ids, duplicates: none
   R2 $ref resolution                                   PASS  29 refs, unresolved: none
   R3 embedded XML well-formed                          PASS  5 XML strings, failures: none
   R4 RFC 3339 timestamps                               PASS  16 timestamps, malformed: none
   R5 EIDO-AgencyRole registry                          PASS  agencies with values outside registry: none
   R6 EIDO-LocationType registry                        PASS  locations with values outside registry: none
   R7 EIDO-IncidentStatus-Common registry               PASS  values outside registry: none
   R8 location by value XOR by reference (2.18)         PASS  violations: none
   R9 additional data by value XOR by reference (2.19)  PASS  violations: none
   R10 at most one owning agency (2.6)                  PASS  owners: ['psap-b.example.org']
   R11 EIDO-IncidentType-Common registry (3.2)          PASS  value: 'MVA'
   R12 size vs 10 MiB recipient minimum (STA-024 3.8.1) PASS  10847 bytes serialized
4. Mutation tests (each is a broken copy of the sample; a defect is 'detected' if the schema or a rule reports it):
   M1 remove required issuingElementIdentification      DETECTED  [schema: 1 error(s)]  expected: schema
   M2 eidoVersion 2.0                                   DETECTED  [schema: 1 error(s)]  expected: schema (enum)
   M3 remove required agencyType                        DETECTED  [schema: 1 error(s)]  expected: schema
   M4 agency role outside registry                      DETECTED  [rules: R5]  expected: rule R5 (schema does not check registry
values)
   M5 malformed top-level timestamp                     DETECTED  [rules: R4]  expected: rule R4 (schema format: date-time is not
enforced by default)
   M6 location with both by-value and by-reference      DETECTED  [rules: R8]  expected: rule R8 (schema does not enforce XOR)
   M7 dangling $ref                                     DETECTED  [rules: R2]  expected: rule R2
   M8 two owning agencies                               DETECTED  [rules: R10]  expected: rule R10 (schema does not enforce single
owner)
   M9 remove required callStateRegistryText             DETECTED  [schema: 1 error(s)]  expected: schema
```

```
   M10 incident type outside registry                DETECTED  [rules: R11]  expected: rule R11 (schema does not check registry
values)

Scope statement: this is a static check of published files on a single computer. No live system, network transport, authentication,
or agency data was involved. It is not an execution of the test cases in the specification.
```

Appendix B. Fixtures

B.1 Differences between fixtures

From	To	JSON path	Change	Reason
TD-01	TD-02R	EIDO.lastUpdateTimeStamp	2021-04-30T14:43:49.439-04:00 to 2021-04-30T14:45:00.000-04:00	Later than every component; fixture time, not delivery time
TD-01	TD-02R	agencyComponent[0] (state.pa.us)	incidentOwningAgencyIndicator true added	Sender owns the incident at S2
TD-01	TD-02R	agencyComponent[1] (psap-b.example.org)	Added: role TransferredTo, agencyType psap, incidentOwningAgencyIndicator false, updatedByAgencyReference state.pa.us	Receiving agency named in the request; synthetic
TD-01	TD-02R	locationComponent[2]	Added CurrentIncident location with new $id; PIDF-LO copied from the RoutingLocation entry	Scene location required for dispatch; address not verified
TD-01	TD-02R	incidentComponent	Added: $id, lastUpdateTimeStamp 14:44:00, updatedByAgencyReference state.pa.us, incidentTypeCommonRegistryText MVA, incidentStatusCommonRegistryText [Active], internalIncidentId TD02-A-0001, locationReference to the CurrentIncident location	Required by STA-024 3.8.1 for a RequestForService
TD-02R	TD-02A	agencyComponent[0].incidentOwningAgencyIndicator	true to false	PSAP A has released ownership
TD-02R	TD-02A	agencyComponent[1].incidentOwningAgencyIndicator	false to true	PSAP B owns the incident after acceptance
TD-02R	TD-02A	agencyComponent[1].updatedByAgencyReference	state.pa.us to psap-b.example.org	PSAP B updated its own component

B.2 TD-02R_request_payload.json (full text)

SHA-256 c04a620e092a66a59d481b9799d8b2358f10b02cc96a41c5809aa4b24a601e58


```
{
 "$id": "urn:emergency:uid:incidentid:a56e556d871:bcf.state.pa.us",
 "lastUpdateTimeStamp": "2021-04-30T14:45:00.000-04:00",
 "updatedByAgencyReference": {
  "$ref": "state.pa.us"
 },
 "updatedByAgentReference": {
  "$ref": "tjones.atroop@state.pa.us"
 },
 "eidoVersion": "1.0",
 "issuingElementIdentification": "idx.state.pa.us",
 "callComponent": [
  {
   "$id": "urn:emergency:uid:callid:a56e556d871:bcf.state.pa.us",
   "lastUpdateTimeStamp": "2021-04-30T14:42:00.0-04:00",
   "updatedByAgencyReference": {
    "$ref": "state.pa.us"
   },
   "updatedByAgentReference": {
    "$ref": "tjones.atroop@state.pa.us"
   },
   "queueIdentifier": "sos@state.pa.us",
   "standardPrimaryCallType": "emergency",
   "direction": "incoming",
   "additionalDataReference": [
    {
     "$ref": "123e4567-e89b-12d3-a456-426614174000"
```

```
    },
    {
     "$ref": "123e4567-e89b-12d3-a456-426614174001"
    },
    {
     "$ref": "123e4567-e89b-12d3-a456-426614174002"
    }
   ],
   "callStartTimestamp": "2021-04-30T14:40:00.0-04:00",
   "answerDate": "2021-04-30T14:42:00.0-04:00",
   "callStateRegistryText": "callAnswered",
   "callBackReference": {
    "$ref": "123e4567-e89b-1234-a456-426614174000"
   },
   "agentReference": [
    {
     "$ref": "tjones.atroop@state.pa.us"
    }
   ],
   "locationReference": [
    {
     "$ref": "123e4567-dead-12d3-a456-426614174000"
    },
    {
     "$ref": "123e4567-beef-12d3-a456-426614174000"
    }
   ],
   "personReference": [
    {
     "$ref": "123e4567-abcd-12d3-a456-426614174000"
    }
   ]
  }
 ],
 "callbackComponent": [
  {
   "$id": "123e4567-e89b-1234-a456-426614174000",
   "lastUpdateTimeStamp": "2021-04-30T14:40:00.0-04:00",
   "updatedByAgencyReference": {
    "$ref": "state.pa.us"
   },
   "updatedByAgentReference": {
    "$ref": "tjones.atroop@state.pa.us"
   },
   "callBackInformationUri": [
    "sip:8195551212@osp.com",
    "tel:8195551212"
   ],
   "deviceContactHeader": "sip:conferenceid3053@bridge.ngcs.com;isfocus"
  }
 ],
 "agencyComponent": [
  {
   "$id": "state.pa.us",
   "lastUpdateTimeStamp": "2021-04-30T14:40:00.0-04:00",
   "updatedByAgencyReference": {
    "$ref": "state.pa.us"
   },
   "agencyRoleDescriptionRegistryText": [
    "CallReceiving"
   ],
   "agencyType": [
    "psap"
   ],
   "incidentOwningAgencyIndicator": true
  },
  {
   "$id": "psap-b.example.org",
   "lastUpdateTimeStamp": "2021-04-30T14:44:00.000-04:00",
   "updatedByAgencyReference": {
    "$ref": "state.pa.us"
   },
   "agencyRoleDescriptionRegistryText": [
    "TransferredTo"
   ],
   "agencyType": [
    "psap"
```

```
   ],
   "incidentOwningAgencyIndicator": false
  }
 ],
 "agentComponent": [
  {
   "$id": "tjones.atroop@state.pa.us",
   "lastUpdateTimeStamp": "2021-04-30T14:42:00.0-04:00",
   "updatedByAgencyReference": {
    "$ref": "state.pa.us"
   },
   "updatedByAgentReference": {
    "$ref": "tjones.atroop@state.pa.us"
   },
   "agentWorkstationPositionIdentification": "5",
   "agentRoleRegistryText": [
    "Call Taking"
   ],
   "agencyReference": {
    "$ref": "state.pa.us"
   }
  }
 ],
 "additionalDataComponent": [
  {
   "$id": "123e4567-e89b-12d3-a456-426614174000",
   "lastUpdateTimeStamp": "2021-04-30T14:40:00.0-04:00",
   "updatedByAgencyReference": {
    "$ref": "state.pa.us"
   },
   "additionalDataByValue": "<?xml version=\"1.0\" encoding=\"UTF-8\"?> <svc:EmergencyCallData.ServiceInfo
xmlns:svc=\"urn:ietf:params:xml:ns:EmergencyCallData:ServiceInfo\"> <svc:DataProviderReference>2468.IBOC.MLTS.1359@example.org
</svc:DataProviderReference> <svc:ServiceEnvironment>Unknown</svc:ServiceEnvironment> <svc:ServiceType>wireless</svc:ServiceType>
<svc:ServiceMobility>Mobile</svc:ServiceMobility> </svc:EmergencyCallData.ServiceInfo>"
  },
  {
   "$id": "123e4567-e89b-12d3-a456-426614174001",
   "lastUpdateTimeStamp": "2021-04-30T14:40:00.0-04:00",
   "updatedByAgencyReference": {
    "$ref": "state.pa.us"
   },
   "additionalDataByValue": "<?xml version=\"1.0\" encoding=\"UTF-8\"?> <ad:EmergencyCallData.ProviderInfo
xmlns:ad=\"urn:ietf:params:xml:ns:EmergencyCallData:ProviderInfo\">
<ad:DataProviderReference>string0987654321@example.org</ad:DataProviderReference> <ad:DataProviderString>Example VoIP
Provider</ad:DataProviderString> <ad:ProviderID>urn:nena:companyid:ID123</ad:ProviderID> <ad:ProviderIDSeries>NENA</ad:ProviderIDSeries>
<ad:TypeOfProvider>Telecom Provider</ad:TypeOfProvider>  <ad:ContactURI>tel:+1-201-555-0123</ad:ContactURI> <ad:Language>en</ad:Language>
<ad:DataProviderContact xmlns=\"urn:ietf:params:xml:ns:vcard-4.0\"> <vcard> <fn><text>Hannes Tschofenig</text></fn> <n>
<surname>Hannes</surname> <given>Tschofenig</given> <additional/> <prefix/> <suffix>Dipl. Ing.</suffix> </n> <bday><date>--0203</date></bday>
<anniversary> <date-time>20090808T1430-0500</date-time> </anniversary> <gender><sex>M</sex></gender> <lang>
<parameters><pref><integer>1</integer></pref> </parameters> <language-tag>de</language-tag> </lang> <lang>
<parameters><pref><integer>2</integer></pref> </parameters> <language-tag>en</language-tag> </lang> <org>
<parameters><type><text>work</text></type> </parameters> <text>Example VoIP Provider</text> </org> <adr> <parameters>
<type><text>work</text></type> <label><text>Hannes Tschofenig Linnoitustie 6 Espoo , Finland 02600</text></label> </parameters> <pobox/>
<ext/> <street>Linnoitustie 6</street> <locality>Espoo</locality> <region>Uusimaa</region> <code>02600</code> <country>Finland</country>
</adr> <tel> <parameters> <type> <text>work</text> <text>voice</text> </type> </parameters> <uri>tel:+358 50 4871445</uri> </tel> <tel>
<parameters> <type> <text>work</text> <text>main-number</text> <text>voice</text> </type> </parameters> <uri>tel:+358 50 5050505</uri> </tel>
<email> <parameters><type><text>work</text></type> </parameters> <text>hannes.tschofenig@nsn.com</text> </email> <geo>
<parameters><type><text>work</text></type> </parameters> <uri>geo:60.210796,24.812924</uri> </geo> <key>
<parameters><type><text>home</text></type> </parameters> <uri>http://www.example.com/key.asc</uri> </key>
<tz><text>Finland/Helsinki</text></tz> <url> <parameters><type><text>home</text></type> </parameters> <uri>http://www.tschofenig.priv.at</uri>
</url> </vcard> </ad:DataProviderContact> </ad:EmergencyCallData.ProviderInfo>"
  },
  {
   "$id": "123e4567-e89b-12d3-a456-426614174002",
   "lastUpdateTimeStamp": "2021-04-30T14:40:00.0-04:00",
   "updatedByAgencyReference": {
    "$ref": "state.pa.us"
   },
   "additionalDataByValue": "<?xml version=\"1.0\" encoding=\"UTF-8\"?> <com:EmergencyCallData.Comment
xmlns:com=\"urn:ietf:params:xml:ns:EmergencyCallData:Comment\">
<com:DataProviderReference>string0987654321@example.org</com:DataProviderReference> <com:Comment xml:lang=\"en\">This is an example
text.</com:Comment> </com:EmergencyCallData.Comment>"
  }
 ],
 "locationComponent": [
  {
   "$id": "123e4567-dead-12d3-a456-426614174000",
```

```
   "lastUpdateTimeStamp": "2021-04-30T14:40:00.0-04:00",
   "updatedByAgencyReference": {
    "$ref": "state.pa.us"
   },
   "locationTypeDescriptionRegistryText": "RoutingLocation",
   "locationByValue": "<?xml version=\"1.0\" encoding=\"UTF-8\"?> <presence xmlns=\"urn:ietf:params:xml:ns:pidf\"
xmlns:gp=\"urn:ietf:params:xml:ns:pidf:geopriv10\" xmlns:gbp=\"urn:ietf:params:xml:ns:pidf:geopriv10:basicPolicy\"
xmlns:dm=\"urn:ietf:params:xml:ns:pidf:data-model\" entity=\"pres:alice@atlanta.example.com\"> <dm:device id=\"target123-1\"> <gp:geopriv>
<gp:location-info> <civicAddress xmlns=\"urn:ietf:params:xml:ns:pidf:geopriv10:civicAddr\"> <country>US</country> <A1>DE</A1>
<A3>Wilmington</A3> <PRD>W</PRD> <RD>Cellular</RD> <STS>Street</STS> <HNO>42</HNO> </civicAddress> </gp:location-info> <gp:usage-rules>
<gbp:retransmission-allowed>true</gbp:retransmission-allowed> <gbp:retention-expiry>2021-12-10T16:00:00-04:00</gbp:retention-expiry>
</gp:usage-rules> <gp:method>Manual</gp:method> <gp:provided-by xmlns=\"urn:ietf:params:xml:ns:EmergencyCallData\"> <EmergencyCallDataValue>
<EmergencyCallData.ProviderInfo xmlns=\"urn:ietf:params:xml:ns:EmergencyCallData:ProviderInfo\">
<DataProviderReference>88QV4FpfZ976T@example.com</DataProviderReference> <DataProviderString>Diamond State Exemplar</DataProviderString>
<ProviderID>urn:nena:companyid:diamond</ProviderID> <ProviderIDSeries>NENA</ProviderIDSeries> <TypeOfProvider>Access Network
Provider</TypeOfProvider> <ContactURI>tel:+1-302-555-0000</ContactURI> <Language>en</Language> </EmergencyCallData.ProviderInfo>
</EmergencyCallDataValue> </gp:provided-by> </gp:geopriv> </dm:device> </presence>"
  },
  {
   "$id": "123e4567-beef-12d3-a456-426614174000",
   "lastUpdateTimeStamp": "2021-04-30T14:40:30.0-04:00",
   "updatedByAgencyReference": {
    "$ref": "state.pa.us"
   },
   "locationTypeDescriptionRegistryText": "Caller",
   "locationByReferenceUrl": "https://lrf.osp.com?oasufdqwerjn12346jfias"
  },
  {
   "$id": "urn:emergency:uid:location:td02-incident-scene:example.org",
   "lastUpdateTimeStamp": "2021-04-30T14:44:00.000-04:00",
   "updatedByAgencyReference": {
    "$ref": "state.pa.us"
   },
   "locationTypeDescriptionRegistryText": "CurrentIncident",
   "locationByValue": "<?xml version=\"1.0\" encoding=\"UTF-8\"?> <presence xmlns=\"urn:ietf:params:xml:ns:pidf\"
xmlns:gp=\"urn:ietf:params:xml:ns:pidf:geopriv10\" xmlns:gbp=\"urn:ietf:params:xml:ns:pidf:geopriv10:basicPolicy\"
xmlns:dm=\"urn:ietf:params:xml:ns:pidf:data-model\" entity=\"pres:alice@atlanta.example.com\"> <dm:device id=\"target123-1\"> <gp:geopriv>
<gp:location-info> <civicAddress xmlns=\"urn:ietf:params:xml:ns:pidf:geopriv10:civicAddr\"> <country>US</country> <A1>DE</A1>
<A3>Wilmington</A3> <PRD>W</PRD> <RD>Cellular</RD> <STS>Street</STS> <HNO>42</HNO> </civicAddress> </gp:location-info> <gp:usage-rules>
<gbp:retransmission-allowed>true</gbp:retransmission-allowed> <gbp:retention-expiry>2021-12-10T16:00:00-04:00</gbp:retention-expiry>
</gp:usage-rules> <gp:method>Manual</gp:method> <gp:provided-by xmlns=\"urn:ietf:params:xml:ns:EmergencyCallData\"> <EmergencyCallDataValue>
<EmergencyCallData.ProviderInfo xmlns=\"urn:ietf:params:xml:ns:EmergencyCallData:ProviderInfo\">
<DataProviderReference>88QV4FpfZ976T@example.com</DataProviderReference> <DataProviderString>Diamond State Exemplar</DataProviderString>
<ProviderID>urn:nena:companyid:diamond</ProviderID> <ProviderIDSeries>NENA</ProviderIDSeries> <TypeOfProvider>Access Network
Provider</TypeOfProvider> <ContactURI>tel:+1-302-555-0000</ContactURI> <Language>en</Language> </EmergencyCallData.ProviderInfo>
</EmergencyCallDataValue> </gp:provided-by> </gp:geopriv> </dm:device> </presence>"
  }
 ],
 "personComponent": [
  {
   "$id": "123e4567-abcd-12d3-a456-426614174000",
   "lastUpdateTimeStamp": "2021-04-30T14:40:00.0-04:00",
   "updatedByAgencyReference": {
    "$ref": "state.pa.us"
   },
   "personIncidentRoleRegistryText": [
    "Caller"
   ],
   "callIdentifier": [
    "urn:emergency:uid:callid:a56e556d871:bcf.state.pa.us"
   ],
   "callBackReference": {
    "$ref": "123e4567-e89b-1234-a456-426614174000"
   }
  }
 ],
 "@context": {
  "@vocab": "./JSON-LD_Contexts/EmergencyIncidentDataObjectType.jsonld"
 },
 "incidentComponent": {
  "$id": "urn:emergency:uid:incidentdata:td02-001:example.org",
  "lastUpdateTimeStamp": "2021-04-30T14:44:00.000-04:00",
  "updatedByAgencyReference": {
   "$ref": "state.pa.us"
  },
  "incidentTypeCommonRegistryText": "MVA",
  "incidentStatusCommonRegistryText": [
```

```
   "Active"
  ],
  "internalIncidentId": "TD02-A-0001",
  "locationReference": [
   {
    "$ref": "urn:emergency:uid:location:td02-incident-scene:example.org"
   }
  ]
 }
}
```

TD-02A differs from TD-02R only in the three values listed in B.1 (SHA-256 c49537b16b1e650d14b3fde55c200b36bc56f22e9fcc01d114540b79a7898ca2). TD-01 is the unaltered repository file (SHA-256 4f98a553e6e095e25c96a377a068c18ea87fe1674c65199f9a2f7afe0d5bf57d).

Appendix C. Element Working Record (35 rows)

Working record from which Section 3.3 was selected. Columns: required status as stated in the standard's text, cardinality, clause, OpenAPI type, schema enforcement, presence in TD-01 and in TD-02R, and the test cases that use the element. Operational mapping notes are kept in the author's worksheet and are not reproduced here.

Element / path	Required (text)	Card.	Clause	OpenAPI type	Schema enforced	TD-01	TD-02R	Cases
EIDO.$id	Required	1..1	2.1, 2.4	EmergencyIncidentDataObjectType + PrologueType	Presence Y	Y	Y	TC-01, TC-06
EIDO.lastUpdateTimeStamp	Required	1..1	2.1, 2.4	EmergencyIncidentDataObjectType + PrologueType	Presence Y	Y	Y	TC-04
EIDO.updatedByAgencyReference	Required	1..1	2.1, 2.4	EmergencyIncidentDataObjectType + PrologueType > ReferenceType	Presence Y	Y	Y	TC-01
EIDO.updatedByAgentReference	Conditional: if an agent created or updated	0..1	2.1	EmergencyIncidentDataObjectType + PrologueType > ReferenceType	N	Y	Y	TC-04
EIDO.eidoVersion	Required	1..1	2.4	EmergencyIncidentDataObjectType	Value Y	Y	Y	TC-01, TC-02
EIDO.issuingElementIdentification	Required	1..1	2.4	EmergencyIncidentDataObjectType	Presence Y	Y	Y	TC-01, TC-02
EIDO.incidentComponent	Optional	0..1	2.4, 2.9	EmergencyIncidentDataObjectType > IncidentInformationType	N	N	Y	TC-01, TC-03, TC-04
incidentComponent.incidentTypeCommonRegistryText	Conditional: required when incidentComponent is present	1..1 (within incident)	2.9, 3.2	IncidentInformationType	Presence Y	N	Y	TC-03
incidentComponent.incidentStatusCommonRegistryText	Optional	0..*	2.9, 3.3	IncidentInformationType	N	N	Y	TC-04, TC-08
incidentComponent.incidentTypeInternalCode	Optional	0..1	2.9	IncidentInformationType	N	N	N	TC-03
incidentComponent.internalIncidentId	Optional	0..1	2.9	IncidentInformationType	N	N	Y	TC-01, TC-06
incidentComponent.locationReference	Optional (required if available)	0..*	2.9	IncidentInformationType > ReferenceType	N	N	Y	TC-05, TC-08
incidentComponent.lastUpdateTimeStamp	Conditional: required when incidentComponent is present	1..1 (within incident)	2.1, 2.9	IncidentInformationType + PrologueType	Presence Y	N	Y	TC-04
EIDO.callComponent	Optional	0..*	2.4, 2.10	EmergencyIncidentDataObjectType > CallInformationType	N	Y	Y	TC-01
callComponent[].$id	Required(within the call component)	1..1 (per call)	2.1, 2.10	CallInformationType + PrologueType	Presence Y	Y	Y	TC-01, TC-06
callComponent[].standardPrimaryCallType	Required(within the call component)	1..1	2.10	CallInformationType	Presence Y	Y	Y	TC-03
callComponent[].direction	Required(within the call component)	1..1	2.10	CallInformationType	Value Y	Y	Y	TC-01
callComponent[].callStartTimestamp	Required(within the call component)	1..1	2.10	CallInformationType	Presence Y	Y	Y	TC-01, TC-07
callComponent[].answerDate	Optional	0..1	2.10	CallInformationType	N	Y	Y	TC-07
callComponent[].callStateRegistryText	Required(within the call component)	1..1	2.10	CallInformationType	Presence Y	Y	Y	TC-04
callComponent[].locationReference	Conditional (if a call location exists)	0..*	2.10	CallInformationType > ReferenceType	N	Y	Y	TC-05
EIDO.locationComponent	Conditional: include if a location is known	0..*	2.4, 2.18	EmergencyIncidentDataObjectType > LocationInformationType	N	Y	Y	TC-05, TC-08
locationComponent[].locationTypeDescriptionRegistryText	Required(within the location component)	1..1	2.18, 3.8	LocationInformationType	Presence Y	Y	Y	TC-05
locationComponent[].locationByValue	Conditional: when by-reference is absent	0..1	2.18	LocationInformationType	N	Y	Y	TC-05
locationComponent[].locationByReferenceUrl	Conditional: when by-value is absent	0..1	2.18	LocationInformationType	N	Y	Y	TC-05
EIDO.agencyComponent	Required	1..*	2.4, 2.6	EmergencyIncidentDataObjectType > AgencyType	Presence Y	Y	Y	TC-01, TC-08
agencyComponent[].$id	Required(within the agency component)	1..1	2.1, 2.6	AgencyType + PrologueType	Presence Y	Y	Y	TC-01
agencyComponent[].agencyRoleDescriptionRegistryText	Required(within the agency component)	1..*	2.6, 3.1	AgencyType	Presence Y	Y	Y	TC-01, TC-08
agencyComponent[].agencyType	Required(within the agency component)	1..*	2.6	AgencyType	Presence Y	Y	Y	TC-01
agencyComponent[].incidentOwningAgency	Optional	0..1	2.6	AgencyType	N	N	Y	TC-04, TC-08

Indicator								
EIDO.agentComponent	Conditional: if an agent is involved	0..*	2.4, 2.5	EmergencyIncidentDataObjectType > AgentType	N	Y	Y	TC-08
agentComponent[].agencyReference	Required(within the agent component)	1..1	2.5	AgentType > ReferenceType	Presence Y	Y	Y	TC-08
EIDO.additionalDataComponent	Optional	0..*	2.4, 2.19	EmergencyIncidentDataObjectType > AdditionalDataType	N	Y	Y	TC-01
additionalDataComponent[].additionalDataByValue	Conditional: when by-reference is absent	0..1	2.19	AdditionalDataType	N	Y	Y	TC-01
EIDO.emergencyResourceComponent	Optional	0..*	2.4, 2.20	EmergencyIncidentDataObjectType > EmergencyResourceType	N	N	N	

Appendix D. Static Check Script v1.1

D.1 Rules and mutations

Id	Check	Source of the rule	Schema reports it?
Schema	Validation against openapi.yaml, EmergencyIncidentDataObjectType, Draft 7 semantics	openapi.yaml 1.0.1	Yes
R1	No duplicate $id values within the EIDO	STA-021 2.1 (identifiers)	No
R2	Every $ref resolves to a $id in the same EIDO	STA-021 2.1 (ReferenceType)	No
R3	Embedded XML strings (PIDF-LO, EmergencyCallData) are well formed	STA-021 2.18, 2.19	No
R4	Timestamps in RFC 3339 [5] date-time form	STA-021 2.1; STA-010	No (format not enforced by default)
R5	agencyRoleDescriptionRegistryText values in EIDO-AgencyRole	STA-021 3.1	No
R6	locationTypeDescriptionRegistryText values in EIDO-LocationType	STA-021 3.8	No
R7	incidentStatusCommonRegistryText values in EIDO-IncidentStatus-Common	STA-021 3.3	No
R8	locationByValue XOR locationByReferenceUrl	STA-021 2.18	No
R9	additionalDataByValue XOR additionalDataByReferenceUrl	STA-021 2.19	No
R10	At most one agency with incidentOwningAgencyIndicator true	STA-021 2.6	No
R11	incidentTypeCommonRegistryText in EIDO-IncidentType-Common (245 codes, APCO 2.103.2-2019 [4])	STA-021 3.2	No
R12	Serialized size not above 10 mebibytes	STA-024 3.8.1 (code 513)	No
M1, M3, M9	Remove a required property (top level, agency, call)	Schema required lists	Yes
M2	eidoVersion 2.0	Schema enum	Yes
M4	Agency role outside the registry	R5	No
M5	Malformed top-level timestamp	R4	No
M6	Location with both by-value and by-reference	R8	No
M7	Dangling $ref	R2	No
M8	Two owning agencies	R10	No
M10	Incident type outside the registry	R11	No

D.2 Script source

```
SHA-256 e98536895b8bdcdd8cd6cbdcb22e8072ed486c523dd889a9f4a03217315a696c  eido_static_check_v1_1.py
```

```
#!/usr/bin/env python3
# -*- coding: utf-8 -*-
"""
EIDO static check v1.1 (Level 1, no network, no live system).


v1.1 changes (2026-11-08): R11 EIDO-IncidentType-Common registry (STA-021 section 3.2), M10 mutation,
  R12 size against the 10 MiB minimum recipient limit of STA-024 section 3.8.1 (513 Message Too Large),
  operator taken from --operator plus the OS login name instead of a fixed label.


What it does, on the three published files only:
  1. SHA-256 of each input file (so the record says which files were checked)
  2. JSON parse of the sample
  3. Schema validation of the sample against openapi.yaml -> EmergencyIncidentDataObjectType
  4. $id uniqueness and $ref resolution inside the EIDO
  5. Well-formedness of embedded XML strings (PIDF-LO, EmergencyCallData)
  6. Rules from the standard text that the schema does not enforce:
       - registry values (EIDO-AgencyRole, EIDO-LocationType, EIDO-IncidentStatus-Common)
       - RFC 3339 date-time form of timestamps
       - locationByValue XOR locationByReferenceUrl (NENA-STA-021 section 2.18)
       - additionalDataByValue XOR additionalDataByReferenceUrl (section 2.19)
       - at most one agency with incidentOwningAgencyIndicator = true (section 2.6)
       - incidentTypeCommonRegistryText in the EIDO-IncidentType-Common registry (section 3.2)
       - serialized size against the 10 MiB minimum recipient limit (STA-024 section 3.8.1, code 513)
  7. Mutation tests: deliberately broken copies of the sample, to show which
     defects the schema catches and which need a rule check.


Usage:
```

```
  python3 eido_static_check.py --schema openapi.yaml --sample "Sample call transfer EIDO.json" --operator "Your Name" --out report.txt

Requires: Python 3.9+, packages jsonschema and pyyaml
  pip install jsonschema pyyaml
"""
import argparse, copy, datetime, getpass, hashlib, json, platform, re, sys, warnings
from importlib.metadata import version as _ver
warnings.filterwarnings("ignore", category=DeprecationWarning)
import xml.etree.ElementTree as ET


try:
    import yaml
    import jsonschema
    from jsonschema import Draft7Validator
except ImportError as e:
    sys.exit("Missing package: %s. Run: pip install jsonschema pyyaml" % e.name)


# Registry values copied from NENA-STA-021.1a/1b text, sections 3.1, 3.2, 3.3, 3.8
AGENCY_ROLE = {"Assisting", "CallReceiving", "Dispatched", "Dispatching", "Informational", "TransferredTo"}
LOCATION_TYPE = {"RoutingLocation", "Caller", "CurrentIncident", "InitialIncident", "Investigation",
                 "Other", "Staging", "TowerLocation", "UnitLocation"}
INCIDENT_STATUS = {"Active", "Cancelled", "Closed", "FireUnderControl", "NewLocation", "ReOpened",
                   "ResourcesAssigned", "ResourcesEnroute", "ResourcesOnscene", "StructureCleared"}
INCIDENT_TYPE = {
    '911H', '911OPN', 'ABDOM', 'ABDUCTF', 'ABDUCTNF', 'ABUSE', 'ACCIND', 'ACTSHOOT', 'ADMIN', 'AIR',
    'AIRINC', 'ALIEN', 'ALLERGIC', 'ALMA', 'ALMB', 'ALMH', 'ALMR', 'ALRBA', 'ALRBOX', 'ALRBS',
    'ALRCO', 'ALRCO2', 'ALRDEFIB', 'ALRF', 'ALRGAS', 'ALRHEAT', 'ALRKPAD', 'ALRMED', 'ALRNGAS',
    'ALRPANA', 'ALRPANS', 'ALRPROPN', 'ALRPULL', 'ALRROBA', 'ALRROBS', 'ALRU', 'ALRVEH', 'ALRWTR',
    'ALS', 'ANIMAL', 'ASSAULT', 'ASSIST', 'ATL', 'BACK', 'BARICADE', 'BIO', 'BITE', 'BLDCLPSE',
    'BLEEDING', 'BLS', 'BOMB', 'BRGCLPSE', 'BURGLARY', 'BURN', 'CARDIAC', 'CARJACK', 'CDX',
    'CHEMICAL', 'CHESPAIN', 'CHOKE', 'CITASST', 'CIVDIS', 'CIVIL', 'CNTRFT', 'COURT', 'CRIMPS',
    'CRIMTRAF', 'CRWD', 'CU1', 'CU2', 'CU3', 'CURFEW', 'CUST', 'DEATH', 'DEATHNTF', 'DIABETIC',
    'DISORD', 'DISVEH', 'DMPX', 'DROWN', 'DRUGPAR', 'DRUGS', 'DUI', 'DVA', 'DVNA', 'EDP', 'ELEC',
    'ELECHZD', 'ELUDING', 'ENTRAP', 'ESCAPE', 'EXPLOS', 'EXPOSURE', 'EXTORT', 'EYE', 'FAIRC',
    'FALL', 'FAM', 'FAPT', 'FARSON', 'FAS', 'FASX', 'FBRUSH', 'FELEC', 'FFUEL', 'FIGHT', 'FIREWRKS',
    'FMARINE', 'FORCENTR', 'FOTHER', 'FRAUD', 'FREPORT', 'FSMOKE', 'FSTRU', 'FTRAILER', 'FTRAIN',
    'FTRASH', 'FTRUCK', 'FUNK', 'FVEH', 'GASLEAK', 'GASLEAKI', 'GASLEAKO', 'HARR', 'HATE', 'HAZ',
    'HAZMAT', 'HEADACHE', 'HEMRG', 'HLP1', 'HLP2', 'HOLD', 'HOMICIDE', 'HOSTAGE', 'IMP', 'INDECENT',
    'INFO', 'INTERNET', 'INTOX', 'INVEST', 'JUV', 'LARCENY', 'LIQUOR', 'MAJORINC', 'MAYDAY', 'MCI',
    'MISBOATR', 'MISCON', 'MNTL', 'MTLV', 'MVA', 'MVAHR', 'MVAINJY', 'MVAUNK', 'MVCP', 'NCIC',
    'NCRASLTWF', 'NCRCPRF', 'NCRELEV', 'NCRFACTR', 'NCRMUAIDF', 'NCRNDE', 'NCRNONE', 'NCRTASKFORCE',
    'NCRTESTF', 'NCRTRANS', 'NEGLECT', 'NOISE', 'NONEMERG', 'OD', 'ODOR', 'OPEN', 'ORDERV',
    'PANDEMIC', 'PARKING', 'PERFND', 'PERMISS', 'PERSUSP', 'PERUNCON', 'PERWANT', 'PHANDLE', 'PREG',
    'PROPFND', 'PROPLOST', 'PROPSTLN', 'PROW', 'PUBWKS', 'PURSUIT', 'RADIAT', 'RECKDRV', 'REPO',
    'RESCUE', 'RESIST', 'RESP', 'RESPARR', 'ROBBERY', 'RUNAWAY', 'SAR', 'SCH', 'SCRM', 'SEIZURE',
    'SEXOFFNS', 'SHOOT', 'SHOPLIFT', 'SICK', 'SLMP', 'SMOKING', 'SOL', 'STABBING', 'STDBY', 'STKX',
    'STROKE', 'SUBJSTP', 'SUICIDE', 'SUICTHRT', 'SUSP', 'TERROR', 'THAZ', 'THREATS', 'TOW', 'TRAIN',
    'TRAINDRL', 'TRANSFER', 'TRAUMA', 'TRAUMAP', 'TRESPASS', 'TROUBLE', 'TRUANT', 'TSTOP', 'VAG',
    'VANDAL', 'VEHABND', 'VEHASST', 'VEHLOCK', 'VEHREC', 'VEHSTLN', 'VEHSUSP', 'VICE', 'WARRANT',
    'WATER', 'WEAPON', 'WELFARE', 'WILDLIFE', 'WIRES', 'XTRICATE'}
TEN_MIB = 10 * 1024 * 1024  # STA-024 3.8.1: recipient maximum must be >= ten mebibytes
RFC3339 = re.compile(r"^\d{4}-\d{2}-\d{2}T\d{2}:\d{2}:\d{2}(\.\d+)?(Z|[+-]\d{2}:\d{2})$")


def sha256(path):
    h = hashlib.sha256()
    with open(path, "rb") as f:
        h.update(f.read())
    return h.hexdigest()


def load_validator(schema_path):
    with open(schema_path, encoding="utf-8") as f:
        api = yaml.safe_load(f)
    root = {"$ref": "#/components/schemas/EmergencyIncidentDataObjectType", "components": api["components"]}
    # Resolver looks up '#/components/schemas/...' inside this document. External NIEM refs
    # (NIEMComponents/nc.yaml) are only followed if the instance contains ncPersonComponent
    # or ncVehicleComponent; the published sample does not.
    resolver = jsonschema.RefResolver(base_uri="", referrer=root)
    return Draft7Validator(root, resolver=resolver)


def schema_errors(validator, instance):
    errs = sorted(validator.iter_errors(instance), key=lambda e: list(e.absolute_path))
    return ["/".join(str(p) for p in e.absolute_path) + ": " + e.message for e in errs]
```

```
def walk(obj, path=""):
    """yield (path, key, value) for every dict member"""
    if isinstance(obj, dict):
        for k, v in obj.items():
            yield path, k, v
            yield from walk(v, path + "/" + k)
    elif isinstance(obj, list):
        for i, v in enumerate(obj):
            yield from walk(v, path + "[%d]" % i)


def rule_checks(eido):
    out = []
    ids, refs, stamps, xmls = [], [], [], []
    for path, k, v in walk(eido):
        if k == "$id": ids.append((path, v))
        if k == "$ref": refs.append((path, v))
        if k in ("lastUpdateTimeStamp", "callStartTimestamp", "answerDate", "vehicleRelationshipTimeStamp", "eta"):
            stamps.append((path, v))
        if isinstance(v, str) and v.lstrip().startswith("<?xml"):
            xmls.append((path + "/" + k, v))
    idset = [v for _, v in ids]
    dup = {x for x in idset if idset.count(x) > 1}
    out.append(("R1 $id uniqueness", "PASS" if not dup else "FAIL", "%d ids, duplicates: %s" % (len(idset), sorted(dup) or "none")))
    unresolved = [(p, r) for p, r in refs if r not in idset]
    out.append(("R2 $ref resolution", "PASS" if not unresolved else "FAIL", "%d refs, unresolved: %s" % (len(refs), unresolved or "none")))
    bad_x = []
    for p, x in xmls:
        try: ET.fromstring(x)
        except ET.ParseError as e: bad_x.append((p, str(e)))
    out.append(("R3 embedded XML well-formed", "PASS" if not bad_x else "FAIL", "%d XML strings, failures: %s" % (len(xmls), bad_x or
"none")))
    bad_t = [(p, v) for p, v in stamps if not (isinstance(v, str) and RFC3339.match(v))]
    out.append(("R4 RFC 3339 timestamps", "PASS" if not bad_t else "FAIL", "%d timestamps, malformed: %s" % (len(stamps), bad_t or "none")))
    bad_role = [a.get("$id") for a in eido.get("agencyComponent", []) if not set(a.get("agencyRoleDescriptionRegistryText", [])) <=
AGENCY_ROLE]
    out.append(("R5 EIDO-AgencyRole registry", "PASS" if not bad_role else "FAIL", "agencies with values outside registry: %s" % (bad_role or
"none")))
    bad_loc = [l.get("$id") for l in eido.get("locationComponent", []) if l.get("locationTypeDescriptionRegistryText") not in LOCATION_TYPE]
    out.append(("R6 EIDO-LocationType registry", "PASS" if not bad_loc else "FAIL", "locations with values outside registry: %s" % (bad_loc or
"none")))
    inc = eido.get("incidentComponent")
    if inc:
        bad_st = [s for s in inc.get("incidentStatusCommonRegistryText", []) if s not in INCIDENT_STATUS]
        out.append(("R7 EIDO-IncidentStatus-Common registry", "PASS" if not bad_st else "FAIL", "values outside registry: %s" % (bad_st or
"none")))
    else:
        out.append(("R7 EIDO-IncidentStatus-Common registry", "NA", "no incidentComponent in this EIDO"))
    both = [l.get("$id") for l in eido.get("locationComponent", []) if ("locationByValue" in l) == ("locationByReferenceUrl" in l)]
    out.append(("R8 location by value XOR by reference (2.18)", "PASS" if not both else "FAIL", "violations: %s" % (both or "none")))
    both2 = [a.get("$id") for a in eido.get("additionalDataComponent", []) if ("additionalDataByValue" in a) ==
("additionalDataByReferenceUrl" in a)]
    out.append(("R9 additional data by value XOR by reference (2.19)", "PASS" if not both2 else "FAIL", "violations: %s" % (both2 or "none")))
    owners = [a.get("$id") for a in eido.get("agencyComponent", []) if a.get("incidentOwningAgencyIndicator") is True]
    out.append(("R10 at most one owning agency (2.6)", "PASS" if len(owners) <= 1 else "FAIL", "owners: %s" % (owners or "none")))
    if inc and "incidentTypeCommonRegistryText" in inc:
        t = inc.get("incidentTypeCommonRegistryText")
        out.append(("R11 EIDO-IncidentType-Common registry (3.2)", "PASS" if t in INCIDENT_TYPE else "FAIL", "value: %r" % t))
    else:
        out.append(("R11 EIDO-IncidentType-Common registry (3.2)", "NA", "no incidentComponent or no incident type in this EIDO"))
    size = len(json.dumps(eido, ensure_ascii=False).encode("utf-8"))
    out.append(("R12 size vs 10 MiB recipient minimum (STA-024 3.8.1)", "PASS" if size <= TEN_MIB else "FAIL", "%d bytes serialized" % size))
    return out


def mutations(sample):
    """Return list of (name, mutated instance, what should catch it)."""
    m = []
    x = copy.deepcopy(sample); x.pop("issuingElementIdentification", None)
    m.append(("M1 remove required issuingElementIdentification", x, "schema"))
    x = copy.deepcopy(sample); x["eidoVersion"] = "2.0"
    m.append(("M2 eidoVersion 2.0", x, "schema (enum)"))
    x = copy.deepcopy(sample); x["agencyComponent"][0].pop("agencyType", None)
    m.append(("M3 remove required agencyType", x, "schema"))
    x = copy.deepcopy(sample); x["agencyComponent"][0]["agencyRoleDescriptionRegistryText"] = ["Transferring"]
    m.append(("M4 agency role outside registry", x, "rule R5 (schema does not check registry values)"))
    x = copy.deepcopy(sample); x["lastUpdateTimeStamp"] = "04/30/2021 14:43"
    m.append(("M5 malformed top-level timestamp", x, "rule R4 (schema format: date-time is not enforced by default)"))
```

```
    x = copy.deepcopy(sample); x["locationComponent"][0]["locationByReferenceUrl"] = "https://example.invalid/loc"
    m.append(("M6 location with both by-value and by-reference", x, "rule R8 (schema does not enforce XOR)"))
    x = copy.deepcopy(sample); x["callComponent"][0]["agentReference"] = [{"$ref": "nobody@nowhere.example"}]
    m.append(("M7 dangling $ref", x, "rule R2"))
    x = copy.deepcopy(sample)
    a = copy.deepcopy(x["agencyComponent"][0]); a["$id"] = "county-b.example"; a["incidentOwningAgencyIndicator"] = True
    x["agencyComponent"][0]["incidentOwningAgencyIndicator"] = True; x["agencyComponent"].append(a)
    m.append(("M8 two owning agencies", x, "rule R10 (schema does not enforce single owner)"))
    x = copy.deepcopy(sample); x["callComponent"][0].pop("callStateRegistryText", None)
    m.append(("M9 remove required callStateRegistryText", x, "schema"))
    if "incidentComponent" in sample:
        x = copy.deepcopy(sample); x["incidentComponent"]["incidentTypeCommonRegistryText"] = "INVALID_CODE"
        m.append(("M10 incident type outside registry", x, "rule R11 (schema does not check registry values)"))
    return m


def main():
    ap = argparse.ArgumentParser()
    ap.add_argument("--schema", required=True); ap.add_argument("--sample", required=True)
    ap.add_argument("--standard", help="optional: path of the standard .docx, hashed for the record")
    ap.add_argument("--out", default="eido_static_check_report.txt")
    ap.add_argument("--operator", required=True, help="name of the person running this check")
    a = ap.parse_args()
    L = []
    L.append("EIDO static check report (script v1.1)")
    L.append("Run at: " + datetime.datetime.now().astimezone().isoformat(timespec="seconds"))
    L.append("Operator: %s (OS login: %s; host: %s)" % (a.operator, getpass.getuser(), platform.node()))
    L.append("Python %s on %s; jsonschema %s; pyyaml %s" % (platform.python_version(), platform.platform(), _ver("jsonschema"),
_ver("pyyaml")))
    L.append("")
    L.append("Input files (SHA-256):")
    for lab, p in (("schema", a.schema), ("sample", a.sample), ("standard", a.standard)):
        if p: L.append("  %-8s %s  %s" % (lab, sha256(p), p))
    L.append("")
    with open(a.sample, encoding="utf-8") as f:
        sample = json.load(f)
    L.append("1. JSON parse: PASS")
    v = load_validator(a.schema)
    errs = schema_errors(v, sample)
    L.append("2. Schema validation (EmergencyIncidentDataObjectType): %s" % ("PASS, 0 errors" if not errs else "FAIL, %d errors" % len(errs)))
    for e in errs: L.append("     " + e)
    L.append("3. Rule checks on the sample:")
    for name, res, detail in rule_checks(sample):
        L.append("   %-52s %-5s %s" % (name, res, detail))
    L.append("4. Mutation tests (each is a broken copy of the sample; a defect is 'detected' if the schema or a rule reports it):")
    for name, inst, expect in mutations(sample):
        se = schema_errors(v, inst)
        rc = [r for r in rule_checks(inst) if r[1] == "FAIL"]
        detected = bool(se) or bool(rc)
        how = ("schema: %d error(s)" % len(se) if se else "") + (" " if se and rc else "") + ("rules: %s" % ", ".join(r[0].split()[0] for r in
rc) if rc else "")
        L.append("   %-52s %s  [%s]  expected: %s" % (name, "DETECTED" if detected else "NOT DETECTED", how or "none", expect))
    L.append("")
    L.append("Scope statement: this is a static check of published files on a single computer. No live system, network transport,
authentication, or agency data was involved. It is not an execution of the test cases in the specification.")
    report = "\n".join(L)
    with open(a.out, "w", encoding="utf-8") as f: f.write(report + "\n")
    print(report)


if __name__ == "__main__":
    main()
```